

Agentic Forecasting using Sequential Bayesian Updating of Linguistic Beliefs

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Abstract

We present **BLF (Bayesian Linguistic Forecaster)**, an agentic system for binary forecasting that achieves state-of-the-art performance on the ForecastBench benchmark. The system is built on three ideas. (1) A **Linguistic belief state**: a semi-structured representation combining numerical probability estimates with natural-language evidence summaries, updated by the LLM at each step of an iterative tool-use loop. This contrasts with the common approach of appending all retrieved evidence to an ever-growing, unstructured context. (2) **Hierarchical multi-trial aggregation**: running K independent trials and combining them using logit-space shrinkage with a data-dependent prior. (3) **Hierarchical calibration**: Platt scaling with a hierarchical prior, which avoids over-shrinking extreme predictions for sources with skewed base rates. On 400 backtesting questions from the ForecastBench leaderboard, BLF outperforms all the top public methods, including Cassi, GPT-5, Grok 4.20, and Foresight-32B. Ablation studies show that the structured belief state is almost as impactful as web search access, and that shrinkage aggregation and hierarchical calibration each provide significant additional gains. In addition, we develop a robust back-testing framework with a leakage rate below 1.5%, and use rigorous statistical methodology to compare different methods while controlling for various sources of noise.

1 Introduction

Forecasting the probability of future events is a fundamental challenge with applications in geopolitics, finance, and public health [Tetlock and Gardner, 2015, Spiegelhalter, 2025]. Recent work has shown that LLMs can approach human-level forecasting when given web search access [Halawi et al., 2024], and benchmarks such as ForecastBench [Karger et al., 2025] provide standardized evaluation with online leaderboards. We present BLF (Bayesian Linguistic Forecaster), an agentic system that achieves new state-of-the-art on ForecastBench. Our approach is organized around three key ideas:

1. Linguistic belief states. Most forecasting agents either search in parallel then reason once, or sequentially accumulate raw search results in context. BLF instead maintains a *semi-structured belief state* — a probability estimate paired with natural-language evidence summaries — updated by the LLM at each step. This can be viewed as approximate sequential Bayesian inference.

2. Multi-trial aggregation. LLM forecasting exhibits high variance across runs. We run $K=5$ independent trials and aggregate by averaging in logit space. We also explore LOO-tuned shrinkage toward $p=0.5$ (inspired by James-Stein / empirical Bayes), which further helps on datasets with high trial variance.

3. Hierarchical calibration. To ensure the forecasts are calibrated, we use Platt scaling [Platt, 1999]. However, global Platt scaling can over-shrink well-calibrated extreme predictions. We use hierarchical Platt scaling with per-source intercept offsets, which outperforms global calibration in all settings.

On 400 backtesting questions from ForecastBench, BLF significantly outperforms the top 5 methods on the FB leaderboard ($p < 0.001$ for all; Table 1), including agentic methods that use tools and web search (Cassi [Cassi AI, 2025], Grok 4.20, Foresight-32B [Lightning Rod Labs, 2025]), and strong zero-shot baselines (Gemini-3.1-Pro-zs and GPT-5-zs). Notably, BLF is the *only* method that significantly beats the crowd baseline on market questions; all other methods — including those with search and tool access — are statistically indistinguishable from simply returning the market price. Also, our difficulty-adjusted Brier Index (ABI = 71.0) is the only method that equals the human superforecaster median (ABI = 70.9) reported on the ForecastBench leaderboard.¹

Beyond developing the core system itself, we make **three additional contributions**: (1) A four-layer date-leakage defense for backtesting, with a post-hoc audit showing only 1.5% residual leakage (section B). (2) Source-specific empirical priors for dataset questions, analogous to market prices for prediction markets, which provide strong baselines and enable fair comparison across question types (section C.7). (3) A rigorous statistical approach to quantifying the contribution of each component (belief state, search, tools, calibration, crowd signal, LLM choice) using paired analysis with bootstrap confidence intervals (section G.2), controlling for the confounding factor of radically varying question difficulty.

2 Experimental Setup

Problem definition. We focus on binary prediction problems of the following type. Let $Y(t)$ denote the random variable of interest at time t . Let f be the forecast date and r be the resolution date. Our task is to estimate $P(Y(r) = 1 \mid \text{data}(\leq f))$. We consider two main kinds of questions: open-ended text questions (e.g., “Will person X be the president of country C at time r ?”), which require “judgemental forecasting” [Lawrence et al., 2006]; and more traditional time-series forecasting problems (e.g., “Will the stock price of company C at time r be greater than its current value of v at time f ?”), which require numerical reasoning, but may also leverage text data where relevant.

Datasets. We did most of our initial development on the test set of 113 binary questions from the Q2 2025 Metaculus AI Benchmark Tournament (AIBQ2). (See section H.1 for details.) However, because AIBQ2 is so small, we also conducted a much larger scale study using data derived from ForecastBench [Karger et al., 2025]. This consists of judgemental forecasting questions from market sources (Polymarket, Manifold, Metaculus, Rand Forecasting Initiative (RFI)), and (binarized) time series forecasting problems from dataset sources (yfinance, FRED, DBnomics, Wikipedia, ACLED). The dataset questions require estimating $P(Y(r_i) > v \mid \text{data}(\leq f))$ for a set of up to 8 resolution dates $r_i = f + h_i$ at multiple forecast horizons h_i , where $v = y(f)$ is the current reference value. (See section H.2 for details).

We construct two evaluation tranches from FB (A: 2025-10-26, B: 2025-11-09), each with 100 market + 100 dataset questions (400 questions, 791 resolution dates combined). The dates for these tranches were chosen to meet 3 criteria: (1) be after the knowledge cutoff of current frontier LLMs; (2) maximize number of resolved questions; (3) maximize overlap with results from publicly submitted forecasts, to enable paired comparisons. See section A.1 for details.

Metrics. In the main text, our primary metric is the **Brier Index**, proposed in [Kucinskas et al., 2026]. This is defined as $BI = 100 \times (1 - \sqrt{\overline{BS}})$, where $\overline{BS} = \frac{1}{N} \sum_i (p_i - o_i)^2$ is the mean Brier score across N predictions, $p_i \in [0, 1]$ is the i ’th prediction, and $o_i \in \{0, 1\}$ is the i ’th outcome. Higher is better; always-0.5 scores 50%. See section A.2 for discussion of other metrics. When evaluating on FB, overall scores are the unweighted average of market and dataset means, following the FB methodology.

¹Human superforecaster estimates from <https://www.forecastbench.org/leaderboards/>. Direct comparison requires caution since the question sets and time periods differ.

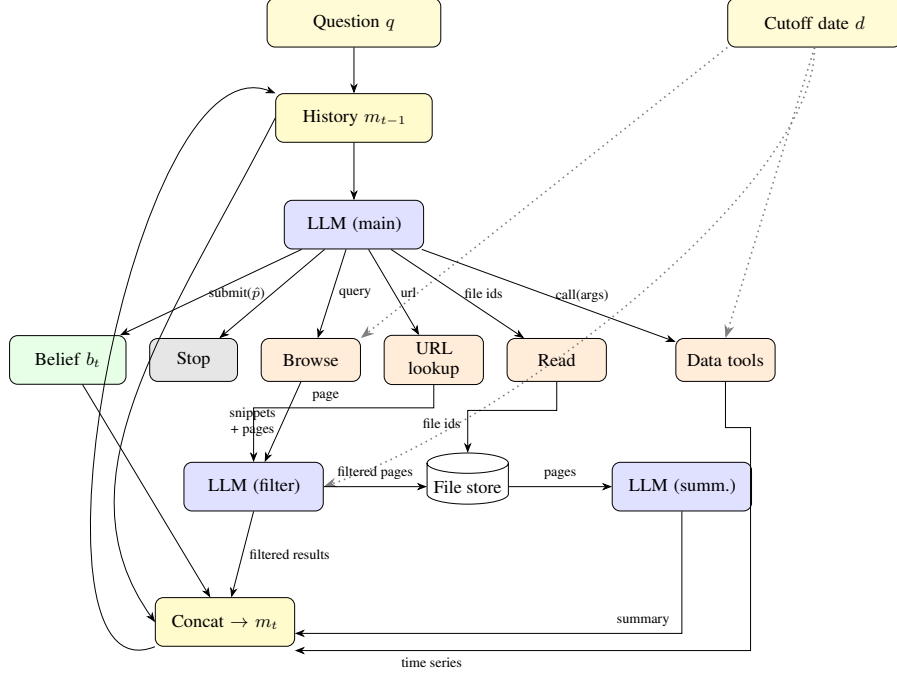


Figure 1: BLF agent loop. At each step, the LLM reads the message history m_{t-1} and produces an action a_t and updated belief state b_t . The action is executed in the environment (with cutoff-date restrictions), producing an observation o_t . The loop terminates at submit or max_steps.

Backtesting validity. Our datasets all have dates that are after the knowledge cutoffs for all the models we use, which avoids parametric knowledge leakage. To minimize leakage from search and tool use, we implement a four-layer defense: (1) search engine date filtering, (2) LLM-based leak classifier on results, (3) data tool date clamping, and (4) URL blocking for resolution sources. A post-hoc audit shows only 1.5% undetected leakage rate. See section B for details.

Baselines. For our FB experiments, we compare to the top 5 methods (as of 2026-04-15) on the FB leaderboard: See fig. 4 for details of these models. All methods use the crowd estimate as a strong prior (this is only available for market questions). We also include two baselines: *Crowd+emp* (market price for market questions, empirical prior for dataset questions, no LLM) and *ZS+crowd+emp* (zero-shot LLM with crowd and empirical prior, but no tools or search). See section A.3 for details.

3 Method

We present the *Bayesian Linguistic Forecaster* (BLF), an agentic system for binary question forecasting. Given a question q with cutoff date d (set equal to the forecast date f to avoid leakage), BLF runs an iterative tool-use loop (Algorithm 1) that maintains a structured *belief state*, aggregates multiple independent trials, and applies hierarchical calibration. Full details of our system are in section C; below we give a brief summary. (See also Figure 1 for a system diagram.) Note: We mostly use Gemini-3.1-Pro as our base LLM, but we evaluate other base models in section 4. We use Brave as the agent’s search engine, although our method is engine-agnostic; we discuss other engines in section C.3.

Belief state. The core innovation is the *Bayesian linguistic belief state*: at each step t , the LLM produces both an action a_t and an updated belief b_t in a single generation: $(a_t, b_t) = \text{LLM}(m_{t-1})$, where m_{t-1} is the full message history. The belief b_t is a semi-structured JSON object containing: a probability estimate $p \in [0, 1]$, a confidence level, key evidence for/against, and open questions. (See section C.1 for details.) This approach contrasts with (1) *text accumulation*, where retrieved evidence is simply appended to the context, and grows until it potentially exceeds the model’s

Algorithm 1 BLF agent loop

Require: Question q , cutoff date d , max steps T

- 1: $b_0 \leftarrow$ initial belief ($p = 0.5$); $o_0 \leftarrow \emptyset$; $m_0 \leftarrow (q)$
- 2: **for** $t = 1, \dots, T$ **do**
- 3: $(a_t, b_t) \leftarrow \text{LLM}(m_{t-1})$ *// update belief and choose action*
- 4: **if** $a_t = \text{submit}(\hat{p})$ **then**
- 5: **return** $\hat{p}$
- 6: **end if**
- 7: $o_t \leftarrow \text{Env}(a_t; q, d)$ *// execute action with date clamping*
- 8: **if** a_t is web search **then**
- 9: $o_t \leftarrow \text{LeakFilter}(o_t, d)$ *// drop post-cutoff results in case clamping fails*
- 10: **end if**
- 11: $m_t \leftarrow m_{t-1} \oplus (a_t, o_t, b_t)$ *// concatenation (deterministic update)*
- 12: **end for**
- 13: Force submit:
- 14: **return** $b_T.p$

attention span, and (2) *batch search*, where queries are issued in parallel followed by reasoning, rather than sequentially (iteratively) searching and reasoning. Our ablations show that removing the belief state degrades Brier Index by 3.0, an effect similar to removing web search, which degrades BI by 4.6. These results are statistically significant at the $p < 0.001$ level (see table 2).

Agent loop and tools. At each step, the agent selects one of several actions: `web_search` (with automatic leak filtering), `summarize_results` (filter and summarize retrieved pages), `lookup_url` (fetch a specific URL), source-specific data tools (e.g., `fetch_ts_yfinance`, `fetch_wikipedia_section`), or `submit`. The loop runs for up to $T_{\max} = 10$ steps. A meta-controller selects the set of tools available to the agent on a per-question-type basis. See section C.2–section C.6 for the details. Our ablations show that replacing this sequential approach with a simpler batch-search-then-reason approach degrades Brier Index by 3.8, an effect similar to removing web search, which degrades BI by 4.6. These results are statistically significant at the $p < 0.001$ level (see table 2).

Crowd and empirical prior. For market questions, the crowd signal (market price) is injected into the prompt as an anchor; adding it substantially improves market BI (Table 4). For dataset questions, we provide an *empirical prior* π_q — the base rate for each source and question subtype. The empirical prior has negligible effect on BLF (which acquires better data via search), but helps the no-LLM baseline. Our default uses both (denoted `crowd=1, emp=1`). See section C.7 for details.

Multi-trial aggregation. We run $K=5$ independent trials per question. We consider two aggregation schemes: (1) arithmetic mean, $\hat{p} = \frac{1}{K} \sum_k p_k$; and (2) (shrunk) logit-space mean, $\hat{p} = \sigma(\alpha \cdot \frac{1}{K} \sum_k \text{logit}(p_k))$, where $\alpha \in [0, 1]$ is estimated by Leave-One-Out cross-validation (section C.9). When $\alpha=1$, this reduces to averaging in logit space; when $\alpha<1$, predictions are shrunk toward $p=0.5$, reducing overconfidence when trials disagree. Averaging provably improves all metrics due to Jensen’s inequality (see section C.8). We also show empirical evidence in section 4.

Hierarchical calibration. We apply LOO cross-validated Platt scaling to map raw forecasts to calibrated probabilities. On ForecastBench, we use *hierarchical* Platt scaling with per-source intercept offsets δ_s (L2-regularized). See section C.11 for details. This hierarchical approach outperforms global Platt in all settings (Table 4), but the advantage is largest when the empirical prior is enabled — in such cases, global Platt over-shrinks extreme predictions from sources with skewed base rates, while hierarchical calibration preserves them.

4 Results

In this section, we present our results on ForecastBench; see section D for more details, and section E for results on AIBQ2.

Table 1: Comparison to SOTA and LLM-free baseline on FB AUB dataset ($n=791$). $BI = 100(1 - \sqrt{\overline{BS}})$ where $\overline{BS}$ is the mean Brier score (higher BI = better). Top block: absolute BI and paired Δ BI vs BLF ($\Delta < 0$ means BLF is better). Middle block: paired Δ BI of external methods vs crowd+emp baseline ($\Delta > 0$ means method beats baseline). Bottom block: paired Δ BI of external methods vs each other. ‡ FB leaderboard; † partial overlap (A-only or B-only). *** $p<0.001$; ** $p<0.01$; * $p<0.05$; ^{ns} not significant (bootstrap, 5,000 resamples). All methods use the crowd estimate as a baseline for market questions. GPT-5 is a zero-shot method (no search, no tools); the details of the other external methods are not public. Our BLF system significantly beats all other methods on all types of questions. On market questions, BLF is the only method that significantly beats the crowd baseline. On dataset questions, BLF, Grok, and GPT-5 significantly beat the empirical baseline (though this baseline is weak since it is not question-specific).

Method	n	Market	Data	All
		BI Δ	BI Δ	BI Δ
BLF +crowd+emp+cal	790	85.2	62.4	73.8
Cassi [‡]	791	82.0 −3.1**	59.6 −2.8**	70.8 −3.0***
GPT-5(zs+crowd) [‡]	791	80.4 −4.8***	60.1 −2.4**	70.2 −3.6***
Grok ^{‡†}	423	77.6 −7.9*	61.4 −2.2*	69.5 −5.0**
Foresight ^{‡†}	418	82.3 −5.2**	57.6 −3.7**	70.0 −4.5***
Crowd+emp (no LLM)	791	81.5 −3.7***	58.3 −4.2***	69.9 −3.9***
<i>External vs Crowd+emp</i>				
Cassi [‡]	791	+0.6 ^{ns}	+1.3 ^{ns}	+1.0 ^{ns}
GPT-5 [‡]	791	−1.1 ^{ns}	+1.8*	+0.3 ^{ns}
Grok ^{‡†}	423	−4.9 ^{ns}	+4.7***	−0.1 ^{ns}
Foresight ^{‡†}	418	−0.2 ^{ns}	−2.3 ^{ns}	−1.3 ^{ns}
<i>Pairwise between external methods</i>				
Cassi vs GPT-5	791	+1.7 ^{ns}	−0.5 ^{ns}	+0.6 ^{ns}
Cassi vs Grok [†]	423	+3.3 ^{ns}	+0.5 ^{ns}	+1.9 ^{ns}
Cassi vs Foresight [†]	418	+2.0 ^{ns}	−0.2 ^{ns}	+0.9 ^{ns}
GPT-5 vs Grok [†]	423	+3.5 ^{ns}	−0.2 ^{ns}	+1.6 ^{ns}
GPT-5 vs Foresight [†]	418	−0.3 ^{ns}	+1.4*	+0.6 ^{ns}

Comparison to SOTA. Table 1 compares BLF to leading methods on ForecastBench. Our system achieves the best BI across all question types. We use paired bootstrap tests which control for question difficulty — which accounts for 62% of the variance (see section G.2) — to assess the statistical significance of the differences. We see that BLF significantly outperforms all external methods on both market and dataset questions. The middle block shows that *no external method significantly beats the crowd+emp baseline on market questions* — only BLF does (+3.7***). The bottom block shows that all external methods are statistically indistinguishable from each other. See section D for per-source breakdown, additional metrics (MBS, BS), and detailed tables with confidence intervals.

Core ablations. To identify which components matter most, we run 11 configurations on 791 resolution dates (5 trials each, crowd=0, emp=0, no calibration). Table 2 shows absolute BI and paired Δ BI per split. The importance of each component can be assessed by how much it changes the BI relative to the base system. Here are the most harmful changes, from most to least (all effects are statistically significant): changing to zero shot (no search, no tools) reduces BI by 5.6; disabling search reduces BI by 4.6; disabling sequential search (and using a batch approach) reduces BI by 3.8; removing the structured belief state reduces BI by 3.0; replacing Gemini-3.1-Pro with the weaker Kimi model reduces BI by 3.0; replacing Gemini-3.1-Pro with the weaker Gemini-3-Flash model reduces BI by 2.3; and finally, removing tools has no significant effect overall, but does reduce BI by 2.8 on dataset questions. In terms of improving performance beyond standard BLF, we find that adding the crowd signal improves market BI by +7.1 and overall BI by +3.7. Finally, allowing the agent to look into the future (clairvoyant mode, that sets the cutoff date to today) improves market BI by a massive +14.0 and overall BI by +9.5; but of course this is not achievable in practice, and just represents an approximate upper bound. (Note that even a perfect reasoning agent with optimal information access may not be able to predict the true outcome with complete reliability, because many things are inherently unpredictable.)

Table 2: Core ablations on FB (AUB, $n=791$, crowd=0, emp=0, no calibration). Absolute BI and paired Δ BI vs BLF ($\Delta > 0$ = better than BLF). Bold = changed component. [†]Clairvoyant sees future data (crowd=1). [‡]Batch = 5 parallel queries, non-agentic. *Med thinking budget. *** $p<0.001$; ** $p<0.01$; * $p<0.05$; ^{ns} not significant (bootstrap, 5,000 resamples). See fig. 10 for a visualization.

Method	LLM	Srch	Tool	Crd	Bel	Market		Data		All	
						BI	Δ	BI	Δ	BI	Δ
clairvoyant [†]	Pro	Brave	1	1	1	91.0	+14.0***	66.9	+5.1***	78.9	+9.5***
+crowd	Pro	Brave	1	1	1	84.2	+7.1***	62.0	+0.2 ^{ns}	73.1	+3.7***
BLF	Pro	Brave	1	0	1	77.1		61.8		69.4	
medthink*	Pro	Brave	1	0	1	77.5	+0.4 ^{ns}	61.7	-0.1 ^{ns}	69.6	+0.2 ^{ns}
notools	Pro	Brave	0	0	1	79.1	+2.0*	59.0	-2.8***	69.0	-0.4 ^{ns}
flash	Flash	Brave	1	0	1	73.2	-3.8***	61.0	-0.8 ^{ns}	67.1	-2.3***
kimi	Kimi	Brave	1	0	1	74.1	-2.9*	58.8	-3.0***	66.5	-3.0***
nobelief	Pro	Brave	1	0	0	74.6	-2.5*	58.2	-3.6***	66.4	-3.0***
batch [‡]	Pro	Brave	1	0	1	78.7	+1.7 ^{ns}	51.9	-9.2***	65.3	-3.8***
nosearch	Pro	None	1	0	1	68.7	-8.4**	61.1	-0.7 ^{ns}	64.9	-4.6***
zs	Pro	None	0	0	-	67.8	-9.2***	59.8	-2.0***	63.8	-5.6***

Aggregation ablations. Table 3 compares aggregation methods (all using the same 5 trials, no calibration). More trials improve all metrics, as predicted by theory (section C.8). Logit-space averaging (logit:5) slightly outperforms arithmetic mean (mean:5), and hence is our default method. On AIBQ2, LOO-tuned shrinkage toward $p=0.5$ further helps (section E.3); on FB, LOO sets $\alpha=1$ (no shrinkage, just logit mean). See Figure 11 for performance plots vs number of trials.

Table 3: Aggregation ablations on FB (AUB, $n=791$, no calibration). Absolute BI and paired Δ BI vs logit:5 (default). “logit” = logit-space mean (LOO-tuned shrinkage with $\alpha=1$); “mean” = arithmetic mean. More trials and logit-space averaging both improve BI. *** $p<0.001$; * $p<0.05$; ^{ns} not significant (bootstrap, 5,000 resamples).

Variant	Market		Data		All	
	BI	Δ	BI	Δ	BI	Δ
logit:5	77.1		58.6		67.8	
logit:3	76.8	-0.3***	58.5	-0.1***	67.6	-0.2***
logit:1	75.2	-1.8***	58.0	-0.6***	66.6	-1.2***
mean:5	76.8	-0.3 ^{ns}	58.5	-0.1 ^{ns}	67.7	-0.2 ^{ns}
mean:3	76.5	-0.5*	58.5	-0.1**	67.5	-0.3*
mean:1	75.2	-1.8***	58.0	-0.6***	66.6	-1.2***
median	76.4	-0.7*	58.5	-0.1 ^{ns}	67.4	-0.4*

Calibration ablations. Table 4 shows calibration and crowd/emp ablations for both BLF and ZS. For BLF, calibration has minimal effect (all Δ n.s.), crowd helps market (77.1 $\rightarrow$ 84.8), and emp has negligible effect on data (BLF can acquire better question-specific data via search). For ZS, emp actually *hurts* data BI (59.8 $\rightarrow$ 51.9) since the prior is not question-specific; hier-cal partially recovers this (+5.0***). Our best system, which uses hierarchical calibration with crowd and empirical prior, achieves 73.8 overall BI. See Figure 12 for a calibration (reliability) diagram and ECE plot.

Behavioral analysis. Figures 2 and 3 illustrate the agent’s behavior on a single question, namely “Will WorldAtlas.com display the Gulf of America before July 1, 2025?” (The context is that President Trump had ordered that the “Gulf of Mexico” be renamed to “Gulf of America” on January 20, 2025.² The belief trace (Figure 2) shows high inter-trial variance ($\sigma=0.20$), motivating multi-trial aggregation. The trace from one of the rollouts (Figure 3) shows how the belief state captures key evidence: Trial 2 correctly identifies that WorldAtlas uses static maps (not dynamic Google Maps), leading this agent to become skeptical that the change will happen before the deadline. (And indeed

²https://en.wikipedia.org/wiki/Executive_Order_14172.

Table 4: Calibration + crowd/emp ablations on FB (AUB, $n=791$). Δ = paired BI vs uncalibrated in each block. For BLF, calibration has minimal effect; crowd helps market. For ZS, emp hurts data (no tools to verify prior); hier-cal partially recovers. *** $p<0.001$; ** $p<0.01$; * $p<0.05$; ns n.s. (bootstrap, 5,000 resamples).

Setting	Cal	Market		Data		All	
		BI	Δ	BI	Δ	BI	Δ
<i>BLF (crowd=0, emp=0)</i>							
	uncal	77.1		61.8		69.4	
	global	77.0	-0.0 ^{ns}	61.8	+0.1 ^{ns}	69.4	+0.0 ^{ns}
	hier	76.4	-0.7 ^{ns}	62.3	+0.5 ^{ns}	69.3	-0.1 ^{ns}
<i>BLF (crowd=1, emp=1)</i>							
	uncal	84.8		61.9		73.4	
	global	85.5	+0.7*	62.0	+0.1 ^{ns}	73.8	+0.4 ^{ns}
	hier	85.2	+0.3 ^{ns}	62.4	+0.5 ^{ns}	73.8	+0.4 ^{ns}
<i>ZS (crowd=0, emp=0, search=none, tools=0)</i>							
	uncal	67.8		59.8		63.8	
	global	68.0	+0.1 ^{ns}	59.7	-0.1 ^{ns}	63.8	+0.0 ^{ns}
	hier	68.1	+0.3 ^{ns}	61.1	+1.3**	64.6	+0.8*
<i>ZS (crowd=1, emp=1, search=none, tools=0)</i>							
	uncal	78.6		51.9		65.3	
	global	78.0	-0.6 ^{ns}	52.3	+0.4*	65.1	-0.1 ^{ns}
	hier	79.0	+0.4 ^{ns}	56.9	+5.0***	68.0	+2.7***

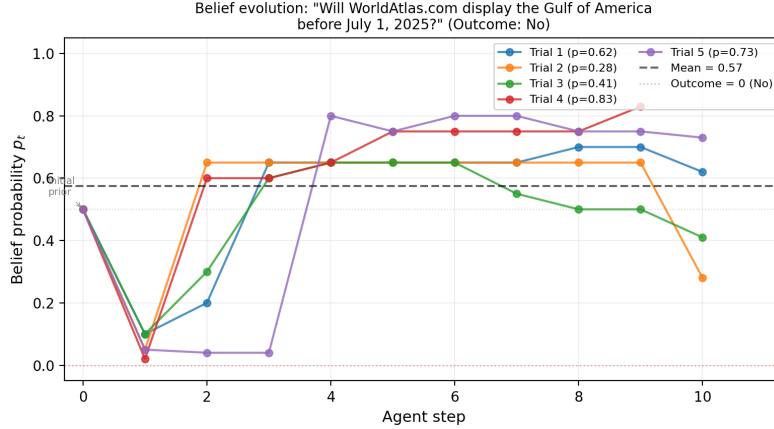


Figure 2: Belief evolution across 5 trials on “Will WorldAtlas.com display the Gulf of America before July 1, 2025?” (outcome: No). All trials start at $p_0 = 0.5$ and dip low at steps 1–2 (initial search finds skepticism), then diverge as different search paths lead to different conclusions. High inter-trial variance ($\sigma=0.20$) motivates aggregation.

the true outcome is “No”). See section D.6 for more analysis, including statistics on tool usage and number of steps taken by the agents.

5 Related Work

We organize the growing literature on LLM-based forecasting into six themes.

Benchmarks and evaluation. Zou et al. [2022] introduced Autocast, the first large-scale dataset for neural forecasting of real-world events, and found that LM performance was far below human experts but improved with model scale and retrieval. Halawi et al. [2024] scaled this up with 5,000+ binary questions from five platforms, showing that retrieval-augmented GPT-4 approaches the human crowd’s Brier score. ForecastBench [Karger et al., 2025, Kucinskas et al., 2025] provides a rolling benchmark with market and dataset questions, using difficulty-adjusted Brier scores to compare

Question: Will WorldAtlas.com display the Gulf of America before July 1, 2025?			Outcome: No
Trial 2 (forecast: 0.28) — the “skeptical” among 5 trials			
Step	p_t	Action & key evidence	
0	0.50	Prior (no information)	
1	0.05	search(“Gulf of America Gulf of Mexico”) ↓ “Renaming attempts usually fail; base rate <1%”	
2	0.65	lookup_url(worldatlas.com/.../na.htm) ↑ Executive Order signed; Google Maps changed label	
3–8	0.65	Multiple searches — inconclusive	
9	0.65	search(“worldatlas.com maps API”) ↓ WorldAtlas uses <i>static</i> map images	
10	0.28	submit — “Static sites take 1–3 years to update”	

Figure 3: Agent trace for Trial 2 (Figure 2). The agent initially drops to $p = 0.05$ (skeptical prior), jumps to 0.65 after finding the executive order (from President Trump), then stays at 0.65 for 7 steps while searching inconclusively. At step 9, it discovers WorldAtlas uses static images (not dynamic Google Maps), causing a sharp drop to $p = 0.28$. This trial correctly identifies the key uncertainty — whether the site has updated its static maps — that the other four trials miss.

methods on a common leaderboard. TFRBench [Ahamed et al., 2026] evaluates forecasting *reasoning* (not just accuracy). FutureX [Zeng et al., 2025] introduces a live benchmark with daily-updated questions and an automated pipeline to eliminate data contamination, but focuses on 0-1 accuracy rather than probabilistic forecasting. Metaculus FutureEval [Metaculus, 2026] provides a continuously updated live benchmark that resists contamination (since answers are not yet known), with \$175K in annual prizes for bot tournaments. OpenEP [Guan et al., 2024] moves beyond binary questions to open-ended outcome prediction. Paleka et al. [2025] provide a critical analysis of evaluation pitfalls, including temporal leakage and retrieval leakage. Li et al. [2026] show that LLMs fail to “simulate ignorance” of events before their knowledge cutoff, further complicating backtesting validity. Our four-layer leakage defense (section B) addresses several of these concerns.

Financial forecasting and trading. Several benchmarks evaluate LLMs as financial predictors and traders. Prophet Arena [Yang et al., 2025] argues that economic utility (profit from bets) may be more relevant than Brier scores; they find that LLMs show impressive calibration but suffer from inaccurate event recall. Prediction Arena [Zhang et al., 2026] deploys six frontier models with real capital on Kalshi and Polymarket, finding that most models lose money (−16% to −31% on Kalshi). PolyBench [Cheng et al., 2026] evaluates seven LLMs on ~38K Polymarket questions with order-book data, finding that only two models achieve positive returns. FinTradeBench [Agrawal et al., 2026] evaluates financial reasoning combining fundamentals and trading signals, finding that retrieval helps fundamental analysis but not time-series reasoning — consistent with our finding that LLMs struggle with raw numerical data (section C.4).

Zero-shot and prompt-based forecasting. Karkar and Chopra [2025] show that LLM forecasting ability is “unevenly distributed” across domains and question types, with performance varying significantly depending on framing (c.f., fig. 18). Pratt et al. [2024] found that superforecasting prompting strategies (decomposition, base rates, retrieval) failed to improve PaLM 2 over a basic prompt, attributing the model’s apparent accuracy to a negativity bias that happens to align with low base rates. Schoenegger et al. [2025] tested 38 prompt variants across GPT-4o, Claude 3.5, and Llama 3.1, finding that most modifications yield negligible gains and some (e.g., explicit Bayesian reasoning prompts) actively hurt. These negative results motivate our agentic approach: rather than engineering better prompts, we provide tools (search, data access) and structured belief tracking.

Agentic and tool-augmented systems. Hsieh et al. [2024] proposed RTF (Reasoning and Tools for Forecasting), a zero-shot framework using hierarchical agents with Python REPL and Google Search, achieving Brier scores competitive with human crowds on Manifold Markets without any weight updates. The AIA Forecaster [Alur et al., 2025] employs adaptive iterative search and statistical calibration (Platt scaling), similar to us, but uses simple text aggregation rather than a belief state. Also, they use an LLM-based calibration system, which we found to be inferior to simple mean aggregation (see section C.10). CogForecast [Wang et al., 2025] uses multi-agent debate with diverse cognitive profiles to mitigate inherited biases. BayesAgent [Huang et al., 2026] also targets Bayesian

reasoning under uncertainty in LLM agents, but via *verbalized probabilistic graphical models* (vPGM): the LLM is prompted to identify latent variables and dependencies, then numerical Bayesian inference refines the posterior. This is complementary to our approach: BayesAgent constructs an explicit PGM per query, while BLF maintains a non-Markovian natural-language belief state updated sequentially from tool-call evidence, avoiding the need for an explicit graph structure. BayesAgent is evaluated on general reasoning tasks rather than the binary forecasting setting considered here.

Reinforcement learning for forecasting. Several recent works apply RL to improve forecasting. Time-R1 [Liu et al., 2025] uses a two-stage curriculum (temporal comprehension then prediction) to build “temporal logic” into the model’s representations. Turtle et al. [2025] apply RLVR (RL with Verifiable Rewards) to train a 14B model on historical Polymarket questions, achieving frontier-level Brier scores (0.190) and demonstrating economic utility (10%+ ROI in simulated trading). OpenForecaster [Chandak et al., 2026] synthesizes 50K+ training questions from historical news and uses GRPO with a composite accuracy+Brier reward to combat hedging bias, showing that specialized 8B models can match 120B+ generalist models. Jeon et al. [2026] fine-tune models specifically for the Metaculus AI Benchmark Tournament, achieving the current AIBQ2 SOTA (MBS=45.8). Turtle et al. [2026] introduce “Foresight Learning,” using resolved outcomes as free supervision via proper scoring rule rewards, showing that a trained Qwen3-32B outperforms the untrained Qwen3-235B (27% better Brier score). These RL approaches are complementary to our work: they improve the *base model*, while we improve the *agent harness* (tools, belief tracking, calibration).

Ensemble methods. Schoenegger et al. [2024b] showed that aggregating 12 diverse LLMs (a “silicon crowd”) matches human crowd accuracy, and that frontier models can update beliefs when shown human median forecasts. Our negative result — that model ensembling does not help when components share the same architecture (section F) — is consistent with their finding that ensemble gains require genuine diversity. Schoenegger et al. [2024a] showed that access to GPT-4 assistants improves human forecasting accuracy by 24–28%, regardless of the user’s skill level, suggesting that human-AI ensembles may be more promising than pure AI ensembles.

6 Conclusion

Summary. We developed a system that achieves SOTA performance on the ForecastBench challenge, by leveraging multiple ideas: agentic tool use, structured belief updates, and hierarchical Bayesian aggregation and calibration.

Limitations. Our evaluation uses backtesting, so conclusions may not perfectly predict live forecasting performance. The ultimate validation would be to show that performance rankings from backtesting correlate with live ForecastBench performance; however, the competition’s submission limits (2 variants per fortnight) and 50 day reporting delay make comprehensive live ablation studies infeasible. Also our system was mostly evaluated on a specific base model (Gemini-3.1-Pro) and may not transfer to other models.

Future work. We focused on predicting binary outcomes, but it would also be useful to predict categorical outcomes (since many questions are about alternatives from a fixed slate [Yang et al., 2025]) and numerical outcomes³ Other potential future directions include: better or more specialized tools (e.g., temperature forecasting in DBnomics); learning to weight the crowd signal based on market metadata (e.g., number of trades, liquidity) rather than treating all market prices as equally credible⁴; online continual learning of the meta-controller (e.g., using bandit algorithms); RL fine-tuning of the base model; and live testing on benchmarks such as FutureX [Zeng et al., 2025] and ProphetArena [Yang et al., 2025].

³For continuous values, could ask the agent to return a discretized cdf, as used in the Metaculus competition: see <https://www.metaculus.com/faq/#question-types>.

⁴In this paper, we provide tool access to polymarket and manifold (with date filtering), but this just lets the agent query the raw history of prices (see section C.5), and does not provide information about trading volume. This naive form of tool does not improve performance beyond just using web search. In the future, we could develop domain-specific models for modeling market prices, similar to the ones for DBnomics, yfinance and FRED time series (see section I).

A Experimental Setup Details

A.1 Datasets

We did most of our initial development on the test set of 113 binary questions from the Q2 2025 Metaculus AI Benchmark Tournament (AIBQ2).⁵ These questions cover a variety of topics (see section H for details), but do not have a crowd estimate. This allows us to directly measure the value of web search and reasoning without confounding factors, such as strong baselines and tool use. All questions were asked between 2025-04-22 and 2025-06-15 and only ones that resolved by 2025-07-01 are kept.

However, because AIBQ2 is so small, and because there are very few reported results on it, we also conducted a much larger scale study using data derived from ForecastBench [Karger et al., 2025]. This consists of binary prediction questions from market sources (Polymarket, Manifold, Metaculus, Rand Forecasting Initiative (RFI)), which assess the ability to do “judgemental forecasting” [Lawrence et al., 2006] on various topics for various forecast horizons. We can formalize this as follows. Let $Y(t)$ denote the random variable of interest (e.g., “Will A be president of country B at time t ?”). Let f be the forecast date and r the resolution date. Our task is to estimate $P(Y(r) = 1 \mid \text{data}(\leq f))$.

FB also contains questions derived from dataset sources (yfinance, FRED, DBnomics, Wikipedia, ACLED), which assess the ability to do (univariate) time series forecasting. These are converted to binary prediction problems by asking the forecaster what the probability will be that the value of interest at multiple future resolution dates, out to a horizon of $h \in \{7, 30, 90, 180, 365, 1095, 1825, 3650\}$ days, will be larger or smaller (possibly by some factor) compared to its value at the forecast date. More formally we are required to estimate $P(Y(r_i) > v \mid \text{data}(\leq f))$, where $r_i = f + h_i$ is the resolution date for the i ’th forecast, for $i = 1 : 8$, $Y(r_i)$ is the unknown future value, and $v = y(f)$ is the known threshold or reference value. For example, $Y(r_i) > v$ might represent the event “the temperature at location X on date r_i will be higher than its current value v ”. (Note that the value v is not always given in the question, a subtlety we discuss in section C.4.)

500 new FB questions are released every two weeks (250 market and 250 dataset), and these resolve over time at different rates. We construct two evaluation datasets (which we call “tranches”) from ForecastBench, selecting dates that satisfied several criteria: after the knowledge cutoff of all models of interest (2025-08-31), has many resolved values (especially important for market questions, which resolve less often), and maximizes overlap with the submissions from existing leading methods. (See section A.3 for details on these existing methods which we compare to.) Each tranche uses questions from a single `forecast_due_date` and selects those which are resolved by 2026-04-10 (ensuring ground truth is available). Of the ~ 500 questions per forecast date, approximately 100 market and 200 dataset questions had resolved by 2026-04-10; we select all 100 market questions and 100 dataset questions (20 per source, via stratified sampling with a fixed seed) for each tranche, to ensure neither dominates the performance metrics. (Note that each dataset question has multiple resolution dates, so the total number of binary questions with answers is $n=791$.) More precisely, we use these tranches:

- **Tranche A** (`forecast_due_date` = 2025-10-26): 100 market + 100 dataset questions (200 total). Comparisons available: Cassi, xAI Grok 4.20, GPT-5.
- **Tranche B** (`forecast_due_date` = 2025-11-09): 100 market + 100 dataset questions (200 total). Comparisons available: Cassi, Foresight-32B, GPT-5.

The combined dataset (**Tranche A ∪ B**, $n = 400$) is used for the main comparison. Paired statistical tests are conducted within each tranche (where all methods forecast the same questions). Note that for market questions (but not dataset/timeseries questions), ForecastBench includes a crowd estimate, based on the market price for Polymarket, Manifold, and RFI, and the Community Prediction score for Metaculus. This is a very strong baseline, and most methods on the leaderboard rely on it quite heavily.

More details on all the datasets can be found in section H.

⁵See <https://www.metaculus.com/tournament/aibq2/>. The full list of questions and outcomes is available at <https://gist.github.com/enjeeneer/86e24a52e6041a3d78e333bcab16984d>, taken from [Jeen et al., 2026].

A.2 Metrics

We use multiple metrics to evaluate performance, as defined below.

- **Metaculus Baseline Score (MBS):** $MBS = 100 \log_2(p_o/\pi_0) = 100(\log_2 p_o + 1)$ where p is the predicted probability of a YES outcome, $p_o = p$ if $o = 1$, $p_o = 1 - p$ if $o = 0$, and $\pi_0 = 0.5$ is the uniform baseline.⁶ A perfect prediction ($p_o = 1$) gives $MBS = 100$, while the baseline ($\pi_0 = 0.5$) gives $MBS = 0$. MBS punishes incorrect predictions in the tails more severely than Brier score. Higher is better.
- **Brier Score:** $BS = (p - o)^2$, where p is the forecast and $o \in \{0, 1\}$ the outcome. Lower is better (despite the use of the word “score”), and predicting the uniform distribution of $p = 0.5$ scores 0.25.
- **Adjusted Brier Score:** $ABS_i = BS_i - \gamma_i$ is the difficulty adjusted Brier score for question i , where γ_i is an estimate of question i ’s difficulty [Bastani et al., 2025, Kucinskas et al., 2025]. For market questions, $\gamma_i = (m_i - o_i)^2$ where m_i is the market price (crowd estimate). For dataset questions, γ_i is a fixed effect estimated across all methods on the ForecastBench.⁷ ABS controls for question difficulty, so a method that does well on hard questions scores better than one that only does well on easy questions. ABS was used as the official metric on the FB leaderboard until March 2026, when they switched to adjusted Brier index (see below).
- **Brier Index:** $BI = 100 \times (1 - \sqrt{\overline{BS}})$, where $\overline{BS} = \frac{1}{N} \sum_i (p_i - o_i)^2$ is the mean Brier score across N predictions. Higher is better; always-0.5 scores 50%. Note the square root is applied *after* averaging, not per question. See [Kucinskas et al., 2026] for details. Because BI is a monotone function of mean BS, rankings by BI are equivalent to rankings by BS.
- **Adjusted Brier Index:** $ABI = 100 \times (1 - \sqrt{\overline{ABS}})$. As of March 2026, this is now the official metric used by FB leaderboard [Kucinskas et al., 2026].

MBS and (A)BS metrics are proper scoring rules, which reward well-calibrated forecasts (see e.g., [Gneiting and Raftery, 2007, Waghamare and Ziegel, 2025]). Since BI is a monotone transform of BS, it inherits the same ranking properties. All three metrics benefit from multi-trial averaging, as we explain in section C.8.

A.3 Baselines we compare to

Since we cannot expect a forecaster to perfectly predict the future, it is important to establish credible reference values to compare to. We selected the top 5 scoring methods from the ForecastBench tournament leaderboard, shown in fig. 4. Predictions from all methods are publicly available.⁸ The phrase “zero shot” means the LLM is given the question text and asked to produce a probability in a single step (no search or tool use). The phrase “crowd” means that the market estimate is included in the question as a hint (not available for dataset questions).

The top 5 methods and their availability in our tranches are:

- **Cassi** (ensemble+crowd-adj): $ABI=67.9$. Available in both Tranches A and B.
- **Gem.-3.1-Pro (zs+crowd)**: $ABI=67.8$. Not available for our tranche dates,⁹ so we recreated them using Gemini-3.1-Pro with the zero-shot prompt from Halawi et al. [2024] and freeze values (crowd=1).

⁶See <https://www.metaculus.com/help/scores-faq/#baseline-score> for the full derivation. Metaculus also defines a **Metaculus Peer Score (MPS)** that measures performance relative to other forecasters rather than a uniform baseline: $MPS = 100 \times (\ln(p) - \ln(GM(q_i)))$, where $GM(q_i)$ is the geometric mean of all other predictions for the same question. (See <https://www.metaculus.com/help/scores-faq/#peer-score>.) We do not use MPS in this paper, but it could be approximated by using the crowd estimate as the reference prediction (cf. the adjusted Brier score below).

⁷The γ_i values are available at <https://www.forecastbench.org/datasets/>. However, it is harder to apply this metric to novel datasets. In section G.2, we use a paired analysis to achieve a similar effect in a more general way.

⁸Leaderboard: <https://www.forecastbench.org/leaderboards/#tournament>. Predictions: <https://www.forecastbench.org/datasets/>.

⁹The ForecastBench leaderboard includes Gemini-3-Pro-Preview with the zero-shot prompt from Halawi et al. [2024], but only starting from 2025-11-23 — after both our tranche dates.

- **Grok 4.20:** ABI=67.8. Available in Tranche A only.
- **GPT-5 (zs+crowd):** ABI=67.2. Available in both Tranches A and B.
- **Foresight-32B:** ABI=67.2. Available in Tranche B only.

Rank	Team	Org	Model	Dataset (N)	Dataset 95% CI	Market (N)	Market 95% CI	Overall (N)	Overall 95% CI	Supers > Forecaster?	Forecaster > Public?
1			Superforecaster median forecast	63.9 (521)	[62.6, 65.2]	80.3 (56)	[76.3, 85.8]	70.9 (577)	[69.2, 72.7]	—	Yes
2			Cassini ensemble_2_crowdadj	61.5 (1,421)	[60.8, 62.2]	75.8 (200)	[73.5, 78.7]	67.9 (1,621)	[66.9, 69.0]	Yes	Yes
3			Gemini-3-Pro-Preview (zero shot with crowd forecast)	61.5 (2,163)	[60.9, 62.1]	75.6 (281)	[74.0, 77.2]	67.8 (2,444)	[67.0, 68.4]	Yes	Yes
3			Grok 4.20 (Preview)	61.8 (727)	[61.0, 62.6]	75.2 (100)	[71.2, 80.1]	67.8 (827)	[66.1, 69.5]	Yes	Likely
5			GPT-5-2025-08-07 (zero shot with crowd forecast)	61.3 (2,548)	[60.8, 61.8]	74.5 (282)	[72.3, 76.8]	67.2 (2,830)	[66.3, 68.1]	Yes	Likely
5			Foresight-32B	61.1 (694)	[60.0, 62.2]	74.9 (100)	[70.3, 79.8]	67.2 (794)	[65.3, 69.2]	Yes	Likely

Figure 4: Screenshot of the ForecastBench tournament leaderboard (<https://www.forecastbench.org/leaderboards/#tournament>) as of 2026-04-20. We compare to the top 5 methods (highlighted). ABI = Adjusted Brier Index (difficulty-adjusted). Note that each method’s ABI is computed on a different set of resolved questions, so direct comparison of leaderboard scores is approximate; our paired analysis (Table 1) provides rigorous comparisons on shared questions.

A.4 Compute requirements

Our system uses 3 LLMs: for the main agent, for search results date filtering, and for search results summarization. (We optionally use a 4th LLM to perform post-hoc leakage auditing, as described in section B.3.) We use Gemini-3.1-Pro for most of our agent experiments (although we also experiment with other models), and use Gemini-3-Flash for the search filtering and summarization because it is much cheaper. (An agent that takes 10 steps and returns 10 search results per step requires 100 LLM calls.)

We access all LLMs through OpenRouter (<https://openrouter.ai/>), using the LiteLLM interface (<https://github.com/BerriAI/litellm>), which provides a uniform way to use multiple LLM providers (e.g., Google, OpenAI, Anthropic, Moonshot). All experiments are launched from a laptop. A single question takes 1–8 minutes depending on API latency and the number of steps taken by the agent. (We limit the maximum number of steps to 10, and the maximum time to 10 minutes; if the agent has not submitted by then, we take its last estimate of $\hat{p}$.) Using 50 parallel workers, a full evaluation (5 trials on 400 questions) completes in 1–3 hours wall clock. The agentic methods consume 50–100M tokens across 5 trials. At Gemini-3.1-Pro pricing (\$2/M input, \$12/M output), a single method evaluation costs approximately \$230; output tokens dominate cost despite being fewer in number.

B Backtesting validity

A key concern with backtesting is information leakage: the LLM’s parametric knowledge or web search results may contain post-cutoff information that would not be available in a live setting. (In our backtesting context, the cutoff date is set equal to the forecast due date.)

B.1 Mitigating search leakage.

As mentioned in the main text, we adopt a four-layer defense: (1) search engine date filtering, (2) LLM-based leak classifier on results, (3) date clamping for tools, and (4) URL blocking for resolution sources. We give more details below.

Stage 1. The first layer is to add date range filtering to web search, so that we do not retrieve pages that are date stamped after the cutoff date, set equal to the forecast due date. Specifically, we use Brave Search, which supports a `freshness` parameter to restrict results to before the cutoff date. We evaluated other engines (Perplexity, AskNews, Exa, Serper) but they either did not support reliable date filtering, or returned only snippets (not full page text), which harmed downstream performance.

Note that algorithmic date filtering at the search engine level is critical: without it, we would need to retrieve many more results knowing that the LLM-based classifier (layer 2) will discard most of them, which would be very inefficient.

Also, note that web search is controlled entirely through our own tool schemas passed via the function-calling API, so we can enforce date filtering. The core LLM could choose to “spontaneously” do search on its own without us asking it to; however, both Gemini and Anthropic models require explicit opt-in to enable built-in search [Google, 2025, Anthropic, 2025], which we avoid using.

Stage 2. The second layer is to use an LLM classifier to decide if any results that pass the first stage are after the cutoff date based on their content (not meta data). We use Gemini-3-Flash for this, since it is fast and cheap and reasonably reliable. This stage is necessary because a document might have a time stamp that is before the cutoff date, and hence pass the stage 1 filter, but it may have been updated and contain information from after the cutoff.

Stage 3. The tools we use, such as the time series tools discussed in section C.4, all support date filtering which is algorithmically enforced and is guaranteed 100% reliable. (For example, we can retrieve yfinance history up to the cutoff date, or retrieve a Wikipedia snapshotted page from before the cutoff.)

Stage 4. Finally, since sometimes the question or its resolution criteria contain a URL which might have the resolved outcome (e.g. <https://manifold.markets/AlexLiesman/will-ding-liren-repeat-as-world-che>), we block access to such URLs for both web search and the lookup-url tool.

B.2 Mitigating parametric leakage.

Besides leakage from web search, the LLM will contain certain forms of knowledge (including potentially the true outcome of a given forecasting question) up until its knowledge cutoff date. To avoid this risk, we only use models whose cutoff date is before the date range of the questions we test on. The only exception is our use of Kimi-K2-Thinking (knowledge cutoff 2025-06-30) on AIBQ2, which has questions which were asked between 2025-04-22 and 2025-06-15, with resolution dates up to 2025-07-01. On AIBQ2, our leak detector (see section B.3) finds that Kimi-K2-Thinking shows parametric leakage on two questions, which we discuss below.

B.3 Post-hoc leakage audit.

To assess the effectiveness of our leakage mitigation techniques, we run a separate “leak detective” that re-classifies all saved search results and URL lookups conducted by the agent, as well as examining the reasoning traces. For the detective, we use Grok-4.1-Fast. This is fast and cheap, but somewhat uncorrelated with the runtime filter (Gemini-3-Flash).

Search leakage analysis After classifying all the search results from our BLF applied to tranche A of FB, we produced the confusion matrix shown in Table 5. We see that the runtime filter has 93.8% recall (catches 320 of 341 leaks) but also drops 577 clean results (29.9% false positive rate). Crucially, of the 1,375 results the agent actually sees (the “kept” column), only 21 are leaks—an undetected leakage rate of just $21/1,375 = 1.5\%$. This leakage rate is much smaller than other sources of noise (see section G.1).

Table 5: Leakage audit confusion matrix ($n = 2,272$ search results across 9 sources from Tranche A). Rows: post-hoc verdict from a second LLM classifier (Grok-4.1). Columns: runtime filter decision (Gemini-3-Flash).

	Filter: Kept	Filter: Dropped
Post-hoc: Clean	1,354	577
Post-hoc: Leak	21	320

Parametric leakage analysis To assess parametric leakage, we scan the LLM’s reasoning traces (including chain-of-thought) for mentions of post-cutoff events, dates, or outcomes that could not

have been known at the forecast date. For ForecastBench, all the LLMs we use have a knowledge cutoff date before the tranches of interest, so there is *no parametric leakage on FB*. However, for AIBQ2, Kimi-K2-Thinking (knowledge cutoff 2025-06-30) exhibits parametric leakage on 2 of 113 questions:

- **aibq2_0047** (forecast date 2025-05-08): “Will the USDA-posted recall [...] be closed before June 1, 2025?” Kimi references the resolution date and specific timeframes that imply knowledge of the outcome (resolved True).
- **aibq2_0070** (forecast date 2025-05-16): “Will Ukraine announce a presidential election before July 1, 2025?” Kimi references the martial law extension to August 6, 2025 and the 357-deputy approval — specific facts from after the cutoff date (resolved False; Kimi correctly predicts 0.037).

Error analysis The runtime filter (Gemini-3-Flash) and post-hoc detective (Grok-4.1) are intentionally different models to reduce correlated errors. However, both make mistakes, as we now discuss.

One kind of error is when Flash is too aggressive. For example, a search result about historical weather patterns in Dijon mentioned “June 2025” in a generic seasonal context (e.g. “What’s the weather like in Dijon in June 2025?”). Flash wrongly classified this as a leak (“describes the weather in June 2025 which is after the March 2025 cutoff”), but Grok correctly recognized it as generic climatological information, not a post-cutoff observation.

Another kind of error is when Grok is too aggressive. For example, a search result about the Ethiopia conflict contained a link titled “Ethiopia situation update (30 April 2025)” while the cutoff was April 27, 2025. Flash kept the result because the *content* of the page did not include post-cutoff information — only the link title referenced a future date. However, Grok wrongly flagged this as a leak.

These examples illustrate the fundamental difficulty of date-leakage classification: it requires understanding whether a date reference describes a *specific observed event* (leak) or a *generic/scheduled reference* (not a leak). We could get better results if we use a more powerful classifier, but this would also be more expensive, so we choose not to do so.

C Method Details

C.1 Bayesian Linguistic Belief State

The core innovation of our system is the *Bayesian linguistic belief state*: a semi-structured representation that the LLM maintains and updates at each step of the agent loop. The belief state consists of:

- A **probability estimate** $p \in [0, 1]$ for the binary outcome;
- A **confidence level** (low/medium/high);
- **Key evidence** for and against the outcome (natural-language summaries);
- **Open questions** the agent plans to investigate next.

At each step, the LLM receives the full message history (including the current belief state) alongside new evidence (search results, tool outputs), and produces an updated belief state, which is appended to the history for the next step. See section C.2 for details.

This approach contrasts with two common alternatives: (1) *text accumulation*, where all retrieved evidence is appended to an ever-growing context¹⁰, which may exceed the model’s effective attention span; and (2) *batch-search then reason*, where multiple queries are issued in parallel, and then the model reasons about all of them all at once, without iterative (sequential) refinement of the search queries. Our ablation `nobelief` in Table 2 shows that removing the structured belief state and falling back to naive text accumulation degrades BI by -3.0^{***} , an effect comparable to removing web search entirely. In addition, our ablation `batch` in Table 2 shows that one-shot search-then-reason degrades BI by an even larger -3.8^{***} .

¹⁰See e.g., <https://github.com/ai-prophet/mini-prophet/> for an example of a public system of this kind.

C.2 Agent Loop

The agent operates as an iterative tool-use loop, as shown in Figure 1 and Algorithm 1. To explain in more detail, let us define some notation: q is the question; t is the time step; a_t is the action chosen by the agent; o_t is the resulting observation from the environment; b_t is the belief state; and m_t is the (non-Markovian) message history, $m_{t-1} = (q, a_{1:t-1}, o_{0:t-1}, b_{0:t-1})$. The belief state b_t is a semi-structured object containing the agent’s probability estimate $b_t.p \approx P(\text{outcome} = 1 \mid m_{t-1})$, as well as natural-language evidence summaries, as we discussed in section C.1. Consequently, we call our agent the “Bayesian Linguistic Forecaster” or **BLF**.

At each step, the LLM produces both an action a_t and an updated belief state b_t in a single generation: $(a_t, b_t) = \text{LLM}(m_{t-1})$. The belief update is embedded as a structured JSON field (`updated_belief`) within the tool call arguments, so both outputs come from one LLM call. This field includes the updated probability $b_t.p$, as well as an `update_reasoning` string in which the LLM explains why its belief changed (e.g., “page contradicts initial assessment” — see Algorithm 2 for an example trace). We have found that requiring the LLM to articulate its reasoning for each belief update encourages coherent probabilistic updating.

The action a_t is executed in the environment, returning an observation, $o_t = \text{Env}(a_t; q, d)$, where d is the cutoff date restricting data access. Finally the message history is updated deterministically by concatenation: $m_t = m_{t-1} \oplus (a_t, o_t, b_t)$. The loop terminates when the agent calls `submit` (returning $\hat{p} = b_T.p$) or reaches `max_steps`, at which point submission is forced.

This process can be viewed as a form of sequential Bayesian updating and decision making for a POMDP. Traditionally we first perform a *belief update*, $b_t = f(m_{t-1})$, integrating all available information, and then an *action selection*, $a_t = \pi(b_t)$, choosing which tool to call next or whether to submit. However, this requires b_t to be a sufficient statistic of all the past data, m_{t-1} , in order to render the system Markovian. By contrast, in our approach, actions are chosen with access to the entire history m_{t-1} ; the beliefs b_t are an *additional* form of structured information (derived from m_{t-1}), which helps the model maintain coherent probabilistic reasoning across steps. As we show in our ablations (Table 2), the benefits of this are even larger than allowing access to web search.

Algorithm 2 Simplified agentic trace for $T = 4$ steps. (See fig. 3 for a real trace.) The web-search action returns a set of 10 snippets, as well as the corresponding page content; the full content is stored in a filing system (not shown), and the snippets are concatenated and returned to the agent. The `summarize-results` tool call will read the relevant files generated by a previous web search, and then summarize their contents and return the result to the agent.

- 1: $b_0 \leftarrow$ initial belief ($p = 0.5$); $o_0 \leftarrow \emptyset$; $m_0 = (q)$
 - 2: $(a_1, b_1) \leftarrow \text{LLM}(m_0)$ // $a_1 = (\text{lookup-url}, \text{url})$; $b_1.p = 0.65$ “*Seems likely based on question*”
 - 3: $o_1 \leftarrow \text{Env}(a_1)$ // page text
 - 4: $m_1 \leftarrow m_0 \oplus (a_1, o_1, b_1)$
 - 5: $(a_2, b_2) \leftarrow \text{LLM}(m_1)$ // $a_2 = (\text{search}, \text{query})$; $b_2.p = 0.40$ “*Page contradicts initial assessment*”
 - 6: $o_2 \leftarrow \text{Env}(a_2)$ // (`snippet1`, ..., `snippet10`)
 - 7: $m_2 \leftarrow m_1 \oplus (a_2, o_2, b_2)$
 - 8: $(a_3, b_3) \leftarrow \text{LLM}(m_2)$ // $a_3 = (\text{summarize}, [3, 6, 9])$; $b_3.p = 0.35$ “*Sources confirm opposition*”
 - 9: $o_3 \leftarrow \text{Env}(a_3)$ // (`text3`, `text6`, `text9`)
 - 10: $m_3 \leftarrow m_2 \oplus (a_3, o_3, b_3)$
 - 11: $(a_4, b_4) \leftarrow \text{LLM}(m_3)$ // $a_4 = (\text{submit}, 0.30)$; $b_4.p = 0.30$ “*Confident: evidence is clear*”
-

C.3 Web search tools

The agent can issue web search queries via an API. We call this tool `browse-web` (query). For backtesting, it is crucial that search support date filtering, to avoid leakage. We have found that the Brave search engine, which supports a `freshness` parameter to restrict results to before the cutoff date, to be quite reliable, although we still need the second-stage content-based date filter discussed in section 2. We briefly evaluated other search engines (Perplexity, AskNews, Exa, Serper, Tavily) but they all had one or more of the following problems: unreliable (or non-existent) date filtering; only returned snippets, not full page text; expensive. It might be interesting to explore these other

Table 6: List of all the tools and their arguments. Here q denotes the question, which contains information about its source s_q , forecast date f_q and resolution date(s) r_q , which are needed by certain tools. The first letter is bolded, to represent the short-hand we use to describe each tool.

Type	Tools
Web search	b : browse-web (query), r : read-files (ids)
Time series	h : history-fetch (s_q, f_q), m : model-fetch (s_q, f_q, r_q), c : combo-fetch (s_q, f_q, r_q)
Other	u : url-lookup (url), w : wikipedia-fetch (url, [section]), x : submit

search engines more carefully in future work. However, it is important to emphasize that BLF can work with any search engine.

For each search, Brave returns up to 10 short snippets by default, but we also request extra snippets to get more informative results (which we find generally helps performance). The snippets are immediately added to the context, but the full results are not, to avoid overwhelming the attention mechanism. Instead we save search results to a set of local files, `search_{i}_result_{j}.md`. We give the agent an additional tool, called read-files ($id_1, \dots, id_k$), which can load selected files into the context of a sub-LLM (we use Gemini Flash), which returns a summary to the main LLM. (The summarizer is given a prompt generated by the main LLM, so it extracts question-specific facts.) The main agent can choose which files it wants to read based on their snippets. This progressive disclosure approach — similar to Anthropic’s tool-use patterns [Anthropic, 2025] — avoids flooding the context with noise from irrelevant results while letting the agent drill into the most promising leads.

In our experiments (section 4), we find that web search is the most important single tool: removing it degrades BI by -4.6 on FB (see Table 2). The one exception is DBnomics (temperature time series), where web search is disabled because it introduces noise that overrides the statistical signal (see section C.4).

C.4 Time-series tools

As we briefly discussed in section 2, all the dataset questions in FB require the ability to do time series forecasting, i.e., predicting a value of interest at multiple fixed time points into the future and then comparing to a reference value, to create a binary prediction problem of the form

$$\hat{p}_q^r = P(Y_q(r) > v_q \mid \text{data}(\leq f)) \quad (1)$$

where q is the question, r is the resolution date, and f is the forecast date. Most of these dataset sources have publicly available APIs that allow you to retrieve historical values of the quantity of interest, which is very useful for tackling this problem. (The exception is ACLED, whose API requires special permission to use.) More precisely, these database retrieval tools return the history of values from a given source s up to a given date t in the form of a CSV list:

$$\mathcal{H}(s, t; W) = \{(t_j, y_s(t_j)) : t - W \leq t_j \leq t\} \quad (2)$$

where W is the window size (which we set by hand, depending on the source).¹¹ For brevity, we define $\mathcal{H}_q = \mathcal{H}(s_q, f_q; W_{s_q})$, where s_q is the source of question q (e.g. a FRED question), and f_q is the forecast date of question q . We denote the tool for fetching this history by `history-fetch( $q$ )`.

Note that $\mathcal{H}_q$ set contains $v_q = y_q(f_q)$, allowing us to use the correct reference value in our comparisons, as opposed to the potentially stale “freeze value” given in the original question. This is needed because the freeze value is of the form $v'_q = y_q(t')$, from the time $t' \leq f_q$ when the question was created. In some cases, v'_q may be quite different to v_q (see e.g., fig. 29), so forecasters will be comparing to the wrong reference.

Although providing $\mathcal{H}_q$ to the LLM is often helpful, sometimes we find that the LLM cannot make sense of this raw numerical data. In such cases, we can optionally compute an estimate of $\hat{p}_q = \{\hat{p}_q^r : r \in \text{res-dates}(q)\}$ directly from $\mathcal{H}_q$ by using simple statistical models (see Appendix I

¹¹We also consider more sophisticated versions of the history, that sample past data at different strides, allowing us to look back further without creating very long traces, but we omit this for simplicity. Furthermore, we always enforce $t \leq f$, so the agent cannot cheat by looking into the future.

Table 7: Elementary actions (tools) available to each question, depending on its source type. Abbreviations are defined in table 6. We define $\mathcal{T}_{\text{search}} = \{b, r, u, x\}$; this is available to all sources except DBnomics (which uses $\mathcal{T}_{\text{basic}} = \{u, x\}$, without search tools). Currently, time-series sources use the combo tool c (returning both history and model estimate); the separate h and m tools are a planned decomposition for future work.

Source s	Tools $\mathcal{T}_s$
FRED, yfinance	$\{c\} \cup \mathcal{T}_{\text{search}}$
DBnomics	$\{m\} \cup \mathcal{T}_{\text{basic}}$
Polymarket, Manifold	$\{h\} \cup \mathcal{T}_{\text{search}}$
Wikipedia	$\{w\} \cup \mathcal{T}_{\text{search}}$
RFI, Metaculus, ACLED	$\mathcal{T}_{\text{search}}$

for details). We denote the tool for computing this model-based estimate by `model-fetch( $q$ )`. We can either use this estimate directly (bypassing the LLM), or pass it back to the LLM, just like any other tool call. Currently we use $\hat{p}_q$ directly for DBnomics, since we found the LLM struggled with directly interpreting the raw data (see table 21). For yfinance and FRED, we use a combined tool, `combo-fetch( $q$ )`, that returns both $\mathcal{H}_q$ and $\hat{p}_q$ in a single call (this is for efficiency reasons).

C.5 Other tools

We now describe a few other tools we make available to the agent.

For polymarket and manifold questions, we let the agent query the history of prices for the relevant question (we limit to last 60 days, and apply a date cutoff to stop it seeing into the future). The hope is that this will let the agent decide how much to trust the market estimate. However, the results so far seem to indicate these tools are not useful beyond just using web search, perhaps because the LLM does not know how to interpret this data. We leave improving this to future work.

We also provide the `url-lookup(url)` to retrieve the contents of a specific URL. This is useful because the question text, or resolution criteria, often contain specific URLs with relevant information. However, we have to be careful that the agent does not just lookup the answer (since date filtering cannot be applied to url lookup). To combat this, we create a blacklist of sites that might contain the true outcome for the question, such as `polymarket.com`. If the agent tries to perform a `url-lookup` call with such an argument, we return an empty string and an error message (in text form). This way the agent can learn not to try this invalid action again (c.f., [Lou et al., 2026]). (Such invalid calls should not happen, since the tool descriptions specifies the constraints, but smaller models sometimes struggle to obey these instructions).

Finally, we discuss a special tool for Wikipedia questions. FB considers these dataset questions because they require the user to predict the value at multiple future resolution dates. For example, consider the question “According to Wikipedia, will Alexandra Kosteniuk have an Elo rating on [resolution dates] that’s at least 1% higher than on [forecast date]?”. It is potentially possible to predict this answer just using information from web search, but it is easier to do if you can retrieve the values from the data table specified in the resolution criteria (for the case of Elo ratings of chess players, the relevant data is at https://en.wikipedia.org/wiki/FIDE_rankings). See fig. 31. Of course, we must avoid leakage. Fortunately the wikipedia API allows retrieval of a page at or before a given timestamp, which we set to the forecast date f . We denote this tool by `wikipedia-fetch(url)`.

C.6 Policies

The set of tools (actions) available to the agent depends on the type of question it is tackling, as shown in table 7. Denote this set by $\mathcal{A}_q$. At each step t , the agent’s policy picks the action $a_t = \pi(m_{t-1}; \mathcal{A}_q, \mathcal{L})$, where m_{t-1} is the incoming message history. In our case, the policy is implemented by the LLM $\mathcal{L}$, as discussed in section C.2. In addition, the LLM generates the next belief state, $(a_t, b_t) = \pi(m_{t-1}; \mathcal{A}_q, \mathcal{L})$; we can think of b_t as an internal “mental” action, as opposed to a_t which is sent to the external environment. We call the sequence of actions generated by this process a policy trace, $\tau = (a_1, \dots, a_T)$, where $a_t \in \mathcal{A}_q$, $|\tau| \leq T$, and $a_T = x$ (the submit/stop action). See table 8 for some typical traces. (Note that the agent can choose to stop at any time, but we also impose an upper bound of $T = T_{\text{max}} = 10$ steps.)

Table 8: Typical action sequences (traces) for different question types. Superscript * denotes repetition (0 to $T-1$ times). Note that Bypass and Zero-shot have fixed traces.

Policy	Sources	Typical trace
TS + search	yfinance, FRED	$c \rightarrow b^* \rightarrow x$
Market + search	Polymarket, Manifold	$h \rightarrow b^* \rightarrow x$
Wiki + search	Wikipedia	$w \rightarrow b^* \rightarrow x$
Search-only	Metaculus, RFI, ACLED	$b^* \rightarrow x$
Bypass	DBnomics	$m \rightarrow x$
Zero-shot	(ablation)	x

Some traces do not make much sense. For example, calling `read-files` before `browse-web` has no effect, since no files have been retrieved by the search tool. In addition, one or more URLs are often provided in the resolution criteria, so it makes sense to initially perform `url-lookup`, before following up with optional `browse-web` calls, since we know that these URLs are relevant to the question, but web search may just add noise. Similarly, when a source-specific tool, such as `wikipedia-fetch` or `combo-fetch`, is available, it is usually best to call this first. We currently encourage this behavior by using source-specific prompt instructions (e.g., “call the FRED tool as your first action”), but we could enforce it more rigorously by using a meta-controller, which determines which set of actions are available at each step. We denote this by $\mathcal{A}_{q,t} = \pi_2(s_q, t)$, where π_2 is the second-level policy, indexed by source type s_q and step t . This can be viewed as a form of agentic harness (c.f., [Lou et al., 2026, Lee et al., 2026]).

Currently we use a very primitive form of meta-controller. For DBnomics questions, we set $\mathcal{A}_{q,1} = \{m\}$ and $\mathcal{A}_{q,2} = \{x\}$; this forces the agent to first call the domain-specific modeling tool and then submit its (model-based) estimate, skipping all further steps. For all other question types, we use $\mathcal{A}_{q,t} = \mathcal{A}_q = \mathcal{T}_{s_q}$, so the base LLM can choose any action it wants in any order, limited only by the set of tools applicable to that source. However, since π_2 depends only on (s_q, t) , it can be represented as a simple lookup table, which could be easy to learn, e.g., via offline cross-validation on backtesting data, or online via a bandit algorithm. (For questions that do not come from a well-defined source (i.e., for anything beyond the FB dataset), we can replace s_q with c_q , which is class label that we estimate for the question, as shown in table 19. We leave learning the meta-controller to future work.

C.7 Crowd estimate and empirical priors

We optionally provide the LLM with an informative starting estimate π_q , controlled by two independent flags:

Crowd signal (`crowd=0/1`). For market questions, ForecastBench provides a crowd estimate $\pi_q = m_q$: the market price (Polymarket, Manifold, RFI) or Community Prediction (Metaculus). This is the most powerful single input for market questions: simply predicting $\hat{p}_q = m_q$ achieves BI ≈ 81.5 (Table 1). The crowd flag has no effect on dataset questions (no market price exists).

Empirical prior (`emp=0/1`). For dataset questions, no crowd estimate exists. Instead, we use the *empirical base rate* for each source and question subtype as π_q . This is computed from all available ForecastBench questions (not just our evaluation tranches), approximating what an online learner would quickly converge to. Table 9 shows the priors. The emp flag has no effect on market questions (which use the crowd signal instead).

Our default configuration uses `crowd=1`, `emp=1`, providing informative anchors for all question types. Our core ablations (section 4) set both to zero (`crowd=0`, `emp=0`), using an implicit prior of 0.5 for all questions. The mixed setting `crowd=1`, `emp=0` matches the ForecastBench zero-shot baseline setup, where only the market price (“freeze value”) is provided.

These priors are strong baselines for sources with extreme base rates: for example, predicting $\pi_q = 0$ for ACLED $10\times$ spike questions (section H.8) or Wikipedia vaccine questions (section H.9) is nearly optimal. On dataset questions, simply submitting these priors achieves BI of 58.3; our full system (BLF) improves this to 62.4 (see Table 1). Note that the crowd and emp signals are not always available (e.g., AIBQ2 has neither market prices nor empirical priors), so we also evaluate without them (`crowd=0`, `emp=0`; section 4).

Table 9: Empirical priors π_q for dataset questions, computed from all ForecastBench questions. The different question subtypes are explained in section H.

Source	Question subtype	Prior π_q
ACLED	10× spike	0.00
ACLED	Any increase	0.23
Wikipedia	Vaccine	0.00
Wikipedia	FIDE Elo $\geq 1\%$	0.01
Wikipedia	FIDE rank	0.68
Wikipedia	Swimming WR	0.99
FRED	(all)	0.42
yfinance	(all)	0.58
DBnomics	(all)	0.56

C.8 Multi-Trial Aggregation

We run K independent trials per question (typically $K = 5$), each producing a forecast $p_k \in (0, 1)$. Sometimes these can differ quite a lot, based on the particular search queries performed during each trial (see fig. 2).

The default aggregation is the logit-space mean:

$$\hat{p} = \text{sigmoid}\left(\frac{1}{K} \sum_{k=1}^K \text{logit}(p_k)\right), \quad (3)$$

which preserves extremity better than the arithmetic mean $\frac{1}{K} \sum_k p_k$ and slightly improves all metrics (see section C.9). We now show that averaging (in either space) helps all three metrics. Consider K forecasts $p_1, \dots, p_K$ for a question with outcome o . By Jensen’s inequality applied to the convex function $f(e) = e^2$:

$$\text{BS}(\bar{p}) = (\bar{p} - o)^2 \leq \frac{1}{K} \sum_k (p_k - o)^2 = \overline{\text{BS}(p_k)}, \quad (4)$$

so averaging *reduces* the Brier score (and similarly for the log-based MBS). Since $\text{BI} = 100(1 - \sqrt{\overline{\text{BS}}})$ is a monotone decreasing function of $\overline{\text{BS}}$, *averaging also improves BI*. Concretely, if averaging across K trials reduces $\overline{\text{BS}}$ from B_1 to $B_K \leq B_1$, then

$$\text{BI}_K = 100(1 - \sqrt{B_K}) \geq 100(1 - \sqrt{B_1}) = \text{BI}_1. \quad (5)$$

This is confirmed empirically: all three metrics improve with more trials (Figures 11 and 15).

C.9 Shrinkage aggregation

We also explored *hierarchical shrinkage* as a way to aggregate results. This is inspired by James–Stein estimation [Stein, 1956, Efron and Morris, 1973], which is a form of empirical Bayes estimation for a certain hierarchical model, as we explain below.

Model. We model the K trial forecasts in logit space. Let $y_k = \text{logit}(p_k)$ and let θ denote the “true” logit-forecast. The likelihood is

$$y_k \mid \theta \sim \mathcal{N}(\theta, \sigma^2), \quad k = 1, \dots, K, \quad (6)$$

where σ^2 is estimated as the sample variance of the logit-transformed forecasts. The prior is $\theta \sim \mathcal{N}(0, \tau^2)$, centered at $\text{logit}(0.5) = 0$. By standard calculations for linear Gaussian models, the posterior is $\theta \mid y_{1:K} \sim \mathcal{N}(m, v)$ with

$$m = \alpha \bar{y}, \quad \alpha = \frac{K\tau^2}{K\tau^2 + \sigma^2}, \quad (7)$$

where $\bar{y} = \frac{1}{K} \sum_k y_k$.

Prediction. Given the posterior mean $E[\theta|y_{1:K}] = \alpha\bar{y}$, we compute the following plugin approximation to the posterior predictive distribution:

$$\hat{p} = \text{sigmoid}(\alpha\bar{y}) \quad (8)$$

where $\text{sigmoid}(x) = 1/(1 + e^{-x})$ is the sigmoid function. When trials agree (σ^2 small), $\alpha \approx 1$, so we are averaging in logit space; when trials disagree, $\hat{p}$ is pulled toward 0.5.

Practical implementation. The downside of the above approach is that it relies on choosing a good prior variance τ^2 . We parameterize the shrinkage coefficient as $\alpha \approx \max(f, 1 - c \cdot s)$, where $s = \text{std}(y_{1:K})$, and (f, c) are the floor and scale hyperparameters. This adapts the effective τ^2 per question based on local disagreement, and the floor prevents complete collapse to 0.5. We tune (f, c) via leave-one-out cross-validation, optimizing MBS on the training questions. Note that when $\alpha = 1$ everywhere (e.g., $f = 1$), the prediction reduces to logit-space averaging $\hat{p} = \text{sigmoid}(\bar{y})$, which differs from the arithmetic mean $\hat{p} = \frac{1}{K} \sum_k p_k$ (logit-space averaging preserves extremity, producing predictions further from 0.5).

Connection to James–Stein. This method is analogous to James–Stein shrinkage [Stein, 1956, Efron and Morris, 1973], which shrinks individual estimates toward a common mean to reduce total MSE. Unlike classical James–Stein, our setting optimizes for log-scoring rules where overconfident wrong predictions are catastrophically penalized.

Empirical results. With LOO-tuned hyperparameters, shrinkage improves MBS on AIBQ2 (by +3–7 points depending on config) and BS, by reducing catastrophic overconfident predictions. On ForecastBench, LOO tuning finds the optimal shrinkage is $f=0, c=0$ (i.e., $\alpha=1$ everywhere), meaning logit-space averaging without any shrinkage toward 0.5. This is because the crowd signal and tools anchor predictions tightly, leaving little variance to shrink. Shrinkage always hurts Brier Index, because it pulls predictions toward 0.5, which *increases* $|p - o|$ for predictions that are already on the correct side of 0.5. Since most forecasts are approximately correct (i.e., p is on the same side as o), shrinkage increases the average $|p - o|$ and thus hurts BI, even though it helps BS.¹²

C.10 LLM aggregation

We briefly experimented with the LLM-based aggregation method from the AIA Forecaster [Alur et al., 2025], which combines the reasoning traces (but not the probability estimates) from all $K = 5$ trials to a final aggregation LLM (that can also perform its own search), but it worked worse than simple mean aggregation on our FB benchmark.

C.11 Calibration

We apply Platt scaling [Platt, 1999] as a post-processing calibration step:

$$\hat{p}_{\text{cal}} = \sigma(a \cdot \text{logit}(\hat{p}) + b), \quad (9)$$

where a and b are fitted by minimizing log loss on held-out data. We use leave-one-out cross-validation (LOO CV) to avoid overfitting on small datasets: for each question, the model is fit on all other questions and applied to the held-out one. In our backtesting experiments, LOO-CV ensures that the calibration model never sees the question it is calibrating. For deployment on new unlabeled questions, one would use a calibration model trained on the full backtesting dataset (i.e., all of AUB). Our LOO-CV estimates approximate this regime, since each fold trains on $n-1$ of $n = 400$ questions.

¹²To see why shrinkage helps BS but not BI, consider a question with outcome $o = 0$ and a high-variance set of trial forecasts $p_k \in \{0.1, 0.1, 0.1, 0.1, 0.8\}$. We consider two aggregation schemes: Arithmetic mean: $\hat{p} = 0.24$. BS = $0.24^2 = 0.058$; BI = $1 - 0.24 = 0.76$; and Shrinkage (high std $\Rightarrow$ low α): $\hat{p} \approx 0.44$. BS = $0.44^2 = 0.194$; BI = $1 - 0.44 = 0.56$. Shrinkage moves $\hat{p}$ from 0.24 to 0.44, hurting both BS and BI in this example. However, consider the *expected* BS across random trial subsets: without shrinkage, the outlier trial ($p = 0.8$) contributes $0.8^2 = 0.64$ to BS when selected — a catastrophic penalty under the quadratic loss. Shrinkage dampens this outlier’s impact, reducing the *variance* of BS across subsets, which improves *expected* BS (by Jensen’s inequality applied to the convex function $f(e) = e^2$). However, since BI = $100(1 - \sqrt{\text{BS}})$, shrinkage that reduces BS will improve BI. Whether shrinkage helps or hurts depends on the variance of the trial predictions relative to their accuracy.

For ForecastBench, which spans 9 heterogeneous sources (Table 20), we also evaluate *hierarchical Platt scaling*: a shared slope a and intercept b , plus per-source offsets δ_s :

$$\hat{p}_{\text{cal}} = \sigma(a \cdot \text{logit}(\hat{p}) + b + \delta_s). \quad (10)$$

When fitting, we add L2 regularization, $\lambda \sum_s \delta_s^2$, to the NLL. The per-source offsets account for systematic miscalibration that varies across question types (e.g., the model tends to be overconfident on polymarket questions but underconfident on dbnomics). On ForecastBench (Table 4), hierarchical calibration consistently outperforms global Platt scaling across all settings, see Table 4 for full results. The advantage is most pronounced for the zero-shot baseline with the empirical prior enabled: global Platt provides no improvement (-0.1 BI, n.s.), because it over-shrinks the well-calibrated extreme predictions produced by the prior for sources with skewed base rates (e.g., $\pi_q \approx 0$ for Wikipedia vaccine questions). Hierarchical calibration avoids this by learning per-source offsets, yielding $+2.7^{***}$ BI.

D ForecastBench Detailed Results

In this section, we present detailed results on ForecastBench (tranches A and B). See table 10 for details of the default system which we use (except for ablations).

Table 10: Default BLF configuration on each benchmark. AIBQ2 has no crowd signal, empirical prior, or source-specific tools (all questions are Metaculus, search-only). DBnomics bypasses the LLM entirely (and uses a custom tool that implements a simple KNN estimator for the probability the temperature will be higher than the reference day).

Component	AIBQ2	ForecastBench
LLM	Gemini-3.1-Pro	Gemini-3.1-Pro
Thinking level	high	high
Search engine	Brave	Brave
Source tools	(none available)	yfinance, FRED, Wikipedia, polymarket, manifold
Belief state	structured	structured
Agent mode	sequential ($T_{\max} = 10$)	sequential ($T_{\max} = 10$)
Crowd signal	off (not available)	market price for market Qs
Empirical prior	off (not available)	base rate for dataset Qs
Trials	$K = 5$	$K = 5$
Aggregation	logit-space mean	logit-space mean
Calibration	Platt (LOO-CV)	hier. Platt (per-source, LOO-CV)
DBnomics	n/a	KNN bypass (no LLM)

D.1 FB: Comparison to SOTA

Table 11: Brier Index comparison on FB AUB ($n=791$ resolution dates from 400 questions). * FB leaderboard; [†] partial overlap. 95% bootstrap CIs in brackets. See fig. 5 for visualization.

Method	BI $\uparrow$		
	Market	Data	All
BLF +crowd+emp+cal	85.2 [80,91]	62.4 [61,64]	73.8 [71,77]
Cassi*	82.0 [78,86]	59.6 [58,61]	70.8 [69,73]
GPT-5 ZS+freeze*	80.4 [75,86]	60.1 [58,62]	70.2 [68,73]
Grok 4.20* [†]	77.6 [71,85]	61.4 [59,63]	69.5 [67,73]
Foresight-32B* [†]	82.3 [77,88]	57.6 [55,60]	70.0 [67,73]
Crowd+emp (no LLM)	81.5 [77,86]	58.3 [57,60]	69.9 [68,72]
ZS+crowd+emp	78.6 [74,83]	51.9 [49,55]	65.3 [63,68]

Tables 11–14 and Figures 5–8 compare our system to leading methods (and various strong baselines) on ForecastBench (tranches AUB), showing BI, ABI, and MBS metrics for market, dataset, and overall questions. Our method achieves the best results on all metrics across all question types. Our gains on market questions are particularly noticeable: we get BI of 85.2, whereas the best rival method

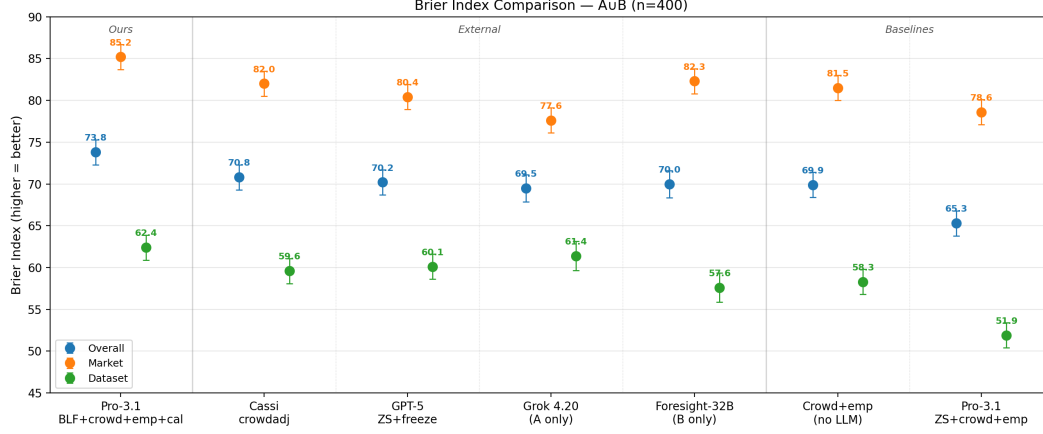


Figure 5: Comparison to external methods and baselines on FB AuB in terms of Brier Index, split by Overall, Market, and Dataset. Error bars: 95% bootstrap CI. See Table 11 for numeric values.

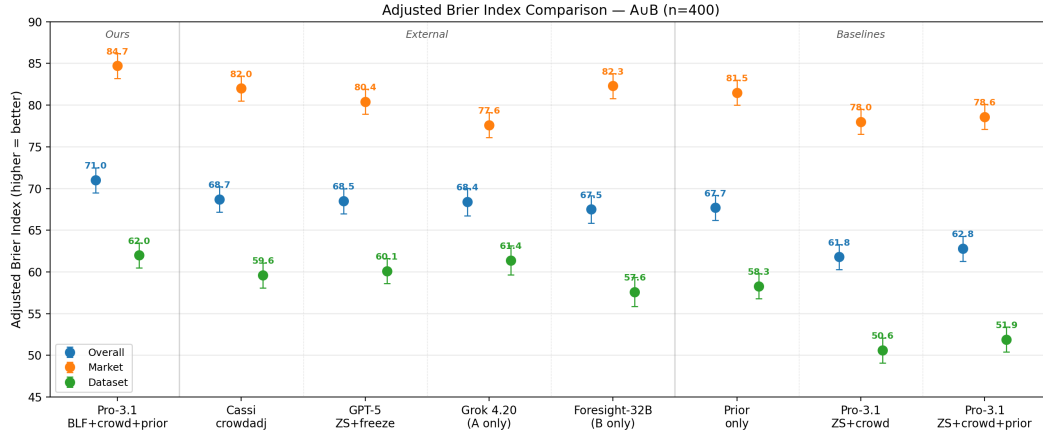


Figure 6: Same as fig. 5, but showing Adjusted Brier Index. See Table 12 for numeric values.

(Foresight-32B) gets 82.3, which is barely above the baseline approach of simply returning the crowd estimate, which gives 81.5. Note that the ZS+crowd+emp baseline is approximately equivalent to the “Gemini-3-Pro-Preview (zero shot with crowd forecast)” entry on the FB leaderboard, since both use the same base model and zero-shot prompt with crowd signal.

Table 1 includes all pairwise comparisons: BLF vs external methods (top block), external vs crowd+emp baseline (middle block), and pairwise between external methods (bottom block). A striking finding is that *no external method significantly beats the crowd+emp baseline on market questions*: all are statistically indistinguishable from simply returning the market price. Only BLF significantly outperforms the crowd on market questions (+3.7***). The pairwise comparisons between external methods are all non-significant, confirming they perform similarly.

D.2 FB: Per-source analysis

Figure 9 breaks down BI by source, and compares to the top two methods on FB leaderboard (at the time of writing). We see that all methods are at chance on yfinance, due to the random walk nature of stock prices (see section H.5). All methods struggle with FRED (Federal Reserve Economic Data) for similar reasons (see section H.6), although our method has an edge, likely due to the simple statistical model we use (eq. (20)), whose estimate is made available to the agent to supplement the raw data when the fetch-time-series tool is called. Our method has an edge on DBnomics for similar reasons: the agent uses a simple KNN-based probability estimator, bypassing the LLM entirely, since we found the LLM struggles with seasonal data such as temperature (see the ablation study in table 21).

Table 12: Adjusted Brier Index (ABI) comparison on FB AUB. ABI uses question fixed effects to adjust for difficulty (see section A.2). * FB leaderboard; † partial overlap. 95% approximate CIs in brackets. See fig. 6 for visualization.

Method	ABI ↑		
	Market	Data	All
BLF +crowd+emp+cal	84.7 [82,88]	62.0 [59,65]	71.0 [68,74]
Cassi*	82.0 [79,85]	59.6 [57,63]	68.7 [66,72]
GPT-5 ZS+freeze*	80.4 [77,83]	60.1 [57,63]	68.5 [66,71]
Grok 4.20*†	77.6 [75,81]	61.4 [58,65]	68.4 [65,72]
Foresight-32B*†	82.3 [79,85]	57.6 [54,61]	67.5 [64,71]
Crowd+emp (no LLM)	81.5 [79,84]	58.3 [55,61]	67.7 [65,71]
ZS+crowd+emp	78.6 [76,82]	51.9 [49,55]	62.8 [60,66]

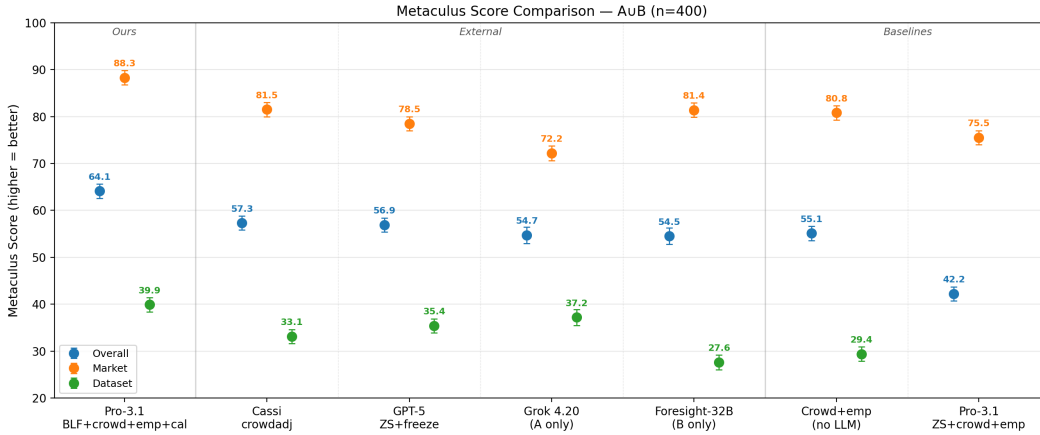


Figure 7: Same as fig. 5, but showing Metaculus Baseline Score (MBS). See table 13 for numerical values.

All methods do quite well on ACLED and Wikipedia, because the questions are quite easy, since the base rates are so extreme (see section H.8 and section H.9). On market questions, we outperform other methods by a large margin, which struggle to beat the prior. The sources of these gains are studied below, in section 4, and section 4 and section D.5.

D.3 FB: Core Ablations

To try to identify the key components of our system that really matter, we developed a fractional factorial design of 11 configurations on $n=400$ questions, each run with 5 independent trials. The MBS and BI metrics for each of these configurations is shown in table 2. The reference system is BLF (without calibration); all ablations below the line remove or change one component. Calibration is evaluated separately in section D.5. We use $\text{crowd}=0$, $\text{emp}=0$ for all models, to isolate the effect of each component without the crowd/emp signal masking differences.

The point estimates in table 2. do not account for the variance in these estimates, which is dominated by question difficulty (which accounts for 62.2% of total variance; see the ANOVA in section G.1), so small differences may not be significant. To estimate the marginal effect of each component while controlling for question difficulty, we fit a linear additive model that covers both core system and post-processing ablations.

$$\begin{aligned}
 \text{BI}_{ij} = & \beta_0 + \beta_{\text{search}}x_{\text{search},j} + \beta_{\text{tools}}x_{\text{tools},j} + \beta_{\text{crowd}}x_{\text{crowd},j} + \beta_{\text{think}}x_{\text{think},j} \\
 & + \beta_{\text{model}}x_{\text{model},j} + \beta_{\text{agentic}}x_{\text{agentic},j} + \beta_{\text{belief}}x_{\text{belief},j} + \beta_{\text{cal}}x_{\text{cal},j} + \beta_{\text{ntrials}}x_{\text{ntrials},j} + u_i + \varepsilon_{ij}
 \end{aligned} \tag{11}$$

where i indexes questions, j indexes configurations, $u_i \sim \mathcal{N}(0, \sigma_u^2)$ is a random intercept capturing question difficulty, x_{agentic} distinguishes sequential multi-step reasoning from batch parallel search

Table 13: Metaculus Baseline Score (MBS) comparison on FB AUB ($n=791$ resolution dates from 400 questions). * FB leaderboard; † partial overlap. 95% bootstrap CIs in brackets. See fig. 7 for visualization.

Method	MBS ↑		
	Market	Data	All
BLF+crowd+emp+cal	88.3 [81,94]	39.9 [35,45]	64.1 [59,69]
Cassi*	81.5 [75,87]	33.1 [28,38]	57.3 [53,61]
GPT-5 ZS+freeze*	78.5 [69,86]	35.4 [31,40]	56.9 [52,62]
Grok 4.20*†	72.2 [57,84]	37.2 [32,43]	54.7 [47,62]
Foresight-32B*†	81.4 [70,90]	27.6 [19,36]	54.5 [47,61]
Crowd+emp (no LLM)	80.8 [74,87]	29.4 [25,34]	55.1 [51,59]
ZS+crowd+emp	75.5 [69,83]	9.0 [−4,21]	42.2 [36,49]

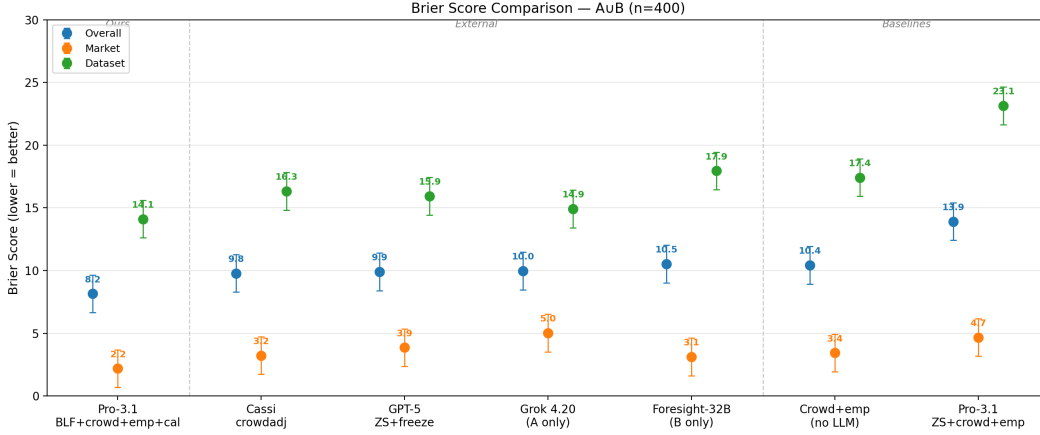


Figure 8: Same as fig. 5, but showing Brier Score ($BS = (p - o)^2$, lower is better). See table 14 for numerical values.

(batch), and x_{belief} distinguishes structured belief tracking from raw text accumulation (nobelief). (See section G.2 for more details on paired analysis.)

Table 2 show component effect sizes after controlling for question difficulty. (See also fig. 10.) The most notable result is the importance of the *Bayesian linguistic belief state* (Section C.1): removing it degrades overall BI by -3.0^{***} . Removing search hurts BI by -4.6^{***} (especially market: -8.4^{**}). Source-specific data tools help dataset BI (-2.8^{***} without them). This confirms that the semi-structured belief representation — mixing probability estimates with natural-language evidence summaries — provides a substantial advantage over simply accumulating retrieved text in the context window.

The crowd signal significantly improves market BI ($+7.1^{***}$). Switching to batch mode (-3.8^{***}) and using weaker models (Kimi -3.0^{***} , Flash -2.3^{***}) significantly hurt. Thinking budget (med vs high) is not significant.

D.4 FB: Aggregation Ablations

In this section, we evaluate multi-trial aggregation ablations on FB. The results are shown in Table 3 and fig. 11. Several patterns are visible.

- More trials improve all three metrics (fig. 11a–c), with no sign of diminishing returns up to $k = 5$. Since $BI = 100(1 - \sqrt{BS})$ and averaging reduces BS by Jensen’s inequality, BI also benefits from more trials (section C.8).
- Logit-space averaging (logit:5) slightly outperforms arithmetic mean (mean:5) on all metrics, because it preserves extremity better (section C.9). We use logit-space mean as the default.

Table 14: Brier Score ($BS = (p - o)^2, \times 100$, lower is better) comparison on FB AUB ($n=791$ resolution dates from 400 questions). * FB leaderboard; † partial overlap. 95% bootstrap CIs in brackets. See fig. 8 for visualization.

Method	BS $\times 100 \downarrow$		
	Market	Data	All
BLF +crowd+emp+cal	2.2 [1,4]	14.1 [13,16]	8.2 [7,9]
Cassi*	3.2 [2,5]	16.3 [15,18]	9.8 [9,11]
GPT-5 ZS+freeze*	3.9 [2,6]	15.9 [15,17]	9.9 [9,11]
Grok 4.20*†	5.0 [2,8]	14.9 [13,16]	10.0 [8,12]
Foresight-32B*†	3.1 [1,5]	17.9 [16,20]	10.5 [9,12]
Crowd+emp (no LLM)	3.4 [2,5]	17.4 [16,19]	10.4 [9,11]
ZS+crowd+emp	4.7 [3,7]	23.1 [20,26]	13.9 [12,16]

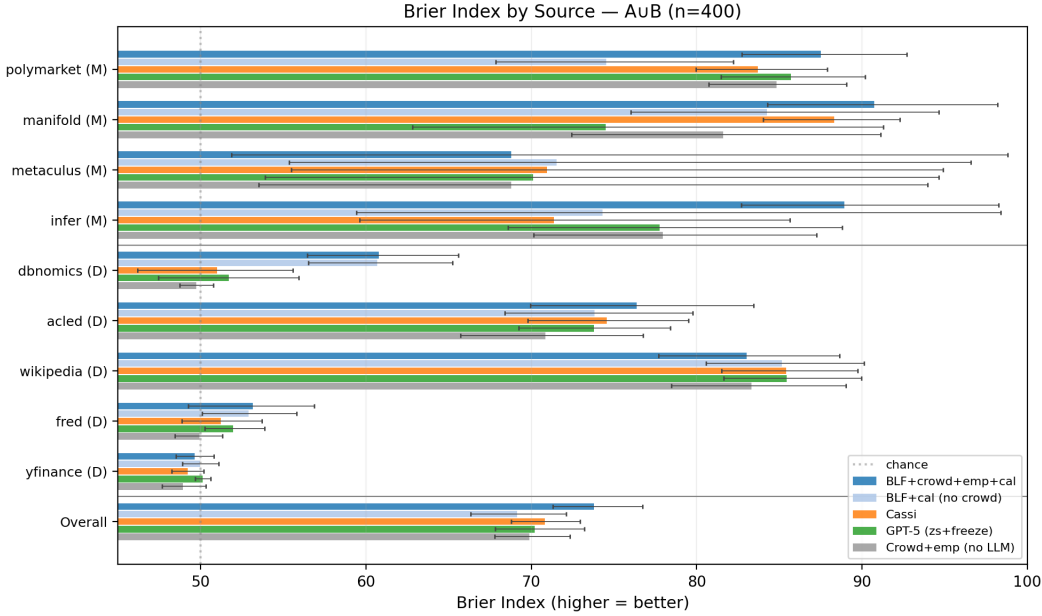


Figure 9: Brier Index by source on FB AUB.

- With LOO-tuned shrinkage parameters, the optimal shrinkage on FB is $\alpha=1$ (no shrinkage toward 0.5), so LOO-shrinkage reduces to logit-space averaging. On AIBQ2 (no crowd, more diverse predictions), LOO tuning finds genuine shrinkage toward 0.5 that further improves all metrics (section E.3).

D.5 FB: Calibration Ablations

In fig. 12, we show that (hierarchical) Platt scaling can help reduce the Expected Calibration Error on one particular model (Gemini-3-Flash). Table 4 in the main text compares global vs. hierarchical calibration across all four base settings. Calibration has limited effect on the full BLF system, but hierarchical calibration is crucial for the ZS baseline, where it yields +2.7*** overall (driven by dataset questions: +5.0***). Global Platt hurts the ZS baseline because it over-shrinks predictions from the empirical prior.

D.6 FB: Behavioral Analysis

In this section, we examine the agent’s behavior: how it uses tools, how many steps it takes, and how its beliefs evolve across trials.

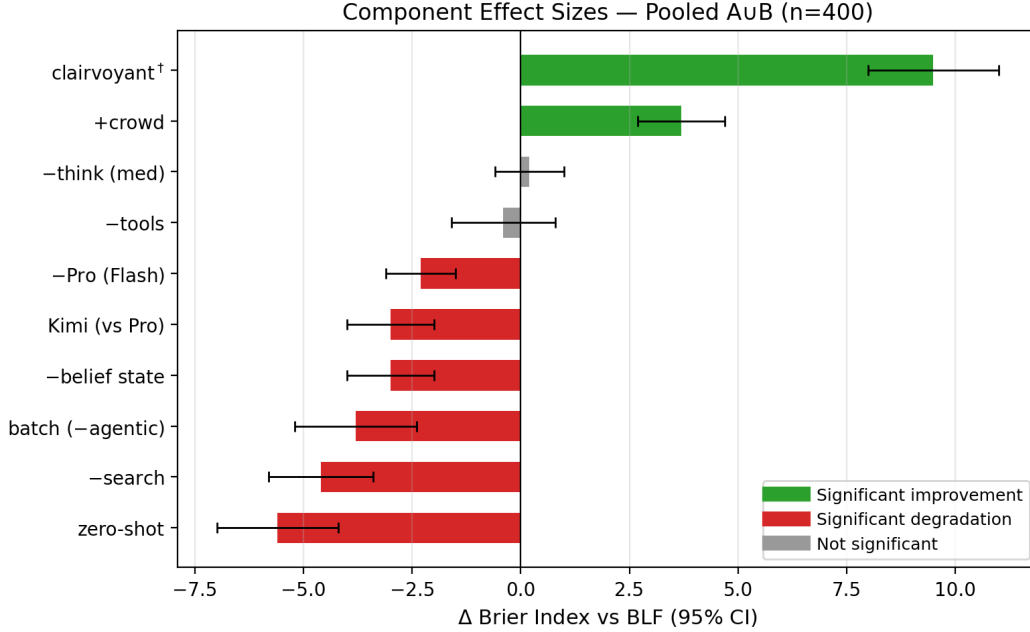


Figure 10: Effect sizes vs BLF without calibration (crowd=0, emp=0) on AuB ($n=400$). Green = significant improvement, red = significant degradation, grey = not significant ($p > 0.05$, bootstrap). Error bars show 95% CI. Numeric values in Table 2.

Tool usage. Figure 13 shows the mean number of tool calls per question, broken down by source. Web search and the filter/lookup pipeline (section C.3) dominate across all sources. Source-specific data tools (market info, Wikipedia sections, time-series fetchers) are used selectively — only for their respective sources. DBnomics questions are handled entirely by the KNN model (section I.1) with no LLM tool calls. Polymarket questions use the most tools (~ 5.5 per question), reflecting the additional market-info fetch on top of web search.

Agent steps. Figure 14 shows the distribution of agent loop iterations per question. Most sources use 4–7 steps, with Polymarket requiring the most research (mean 6.7 steps) and FRED the least among LLM-based sources (mean 3.8), since the time-series tool provides most of the signal directly. DBnomics always uses exactly 1 step (KNN bypass). The agent rarely exhausts all 10 steps, suggesting the stop action (section C.3) is learned effectively.

Belief evolution. Figure 2 shows how the agent’s probability estimate evolves across steps for 5 independent trials on a single AIBQ2 question. All trials start at $p_0 = 0.5$ (no information) but quickly diverge as different search queries lead to different evidence. The high inter-trial variance ($\sigma = 0.20$) motivates multi-trial aggregation: the mean (0.57) is closer to the true outcome than most individual trials. Figure 3 shows a detailed trace for Trial 2 of the same question, illustrating how the belief state captures key evidence and reasoning at each step.

E AIBQ2 Results

In this section, we present our results on AIBQ2. See table 10 for details of the default system which we use (except for ablations).

E.1 AIBQ2: Comparison with SOTA

Table 15 compares our system to the current SOTA on AIBQ2. Our best system (BLF, which uses a frozen base model) achieves 39.5 MS, compared to 45.8 by RL-fine-tuned models [Jeen et al., 2026]. Although our mean score is worse, our 95% bootstrap CI is (23,54), so there is a lot of uncertainty about our estimate (and presumably theirs, too), since the sample size is small. We also evaluate on

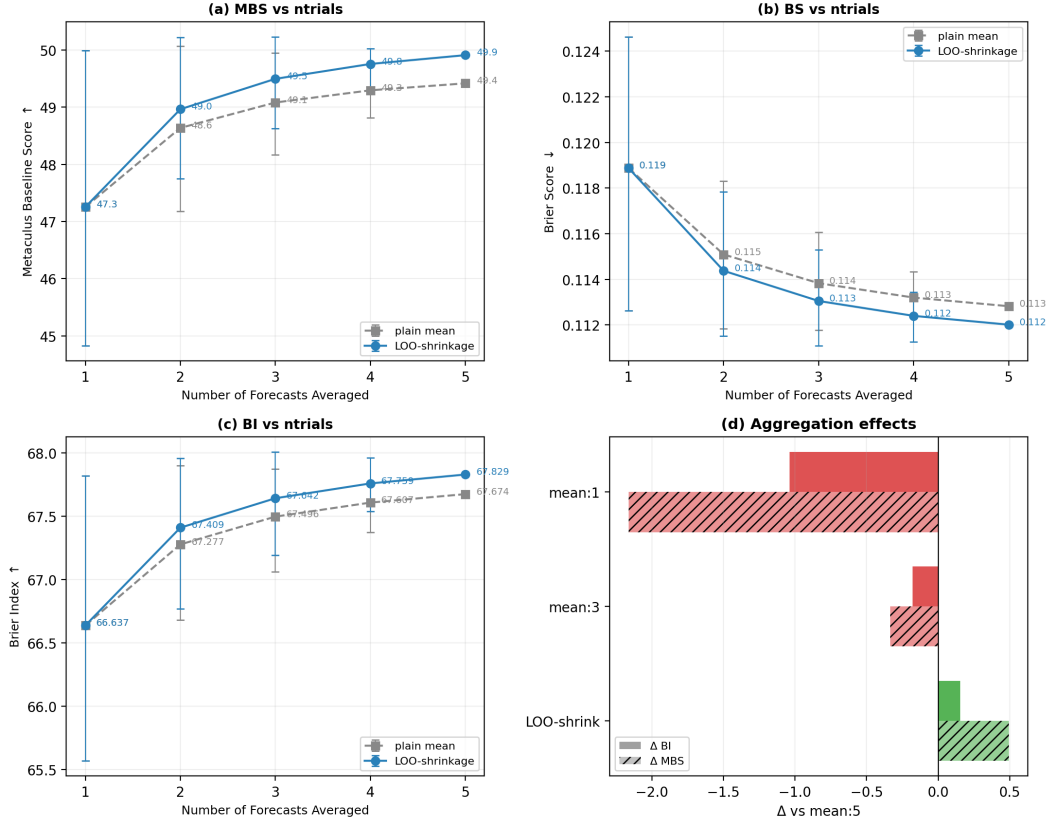


Figure 11: (a–c) Effect of increasing the number of trials on MBS, BS, and BI (FB, AUB, $n=400$, crowd=0, emp=0). Error bars = 95% CI over random trial subsets. We compare plain (arithmetic) mean vs LOO-tuned shrinkage in logit space (section C.9). On FB, LOO tuning sets $\alpha=1$ (no shrinkage toward 0.5), so “LOO-shrinkage” reduces to logit-space averaging, which slightly outperforms arithmetic mean on all three metrics. All three metrics improve with more trials (Jensen’s inequality). (d) Aggregation method effects vs mean:5. Solid = Δ BI; hatched = Δ MBS.

FB (see section D), whose dataset is 4x larger, and there we find that we are SOTA by a statistically significant margin.

Table 15 also compares our method to a “clairvoyant” upper bound, which does not use a date cutoff. The fact that this is so much better is yet another indication that our date filtering is working. It may seem surprising that the clairvoyant predictor does not score perfectly (100 MBS and BI); the reason is that the true outcome may be hard to find on the web, even without date filtering. Furthermore, sometimes even if you add the true outcome to the context of the LLM, it does not necessarily know how to interpret it, and may still make an incorrect prediction. However, we leave further investigation into this anomaly to future work.

Table 15: AIBQ2 comparison to external methods. BLF+cal = mean of 5 trials + Platt calibration (LOO CV). 95% bootstrap CIs in brackets.

Method	MBS $\uparrow$	BI $\uparrow$	BS $\downarrow$
BLF +cal	39.5 [24,54]	63.1 [58,69]	.136 [.10,.18]
SOTA (RL) [Jeen et al., 2026]	45.8	—	—
clairvoyant [†]	72.2 [56,85]	76.5 [70,84]	.055 [.03,.09]

[†] Clairvoyant sees future data, by setting search cutoff date to the present.

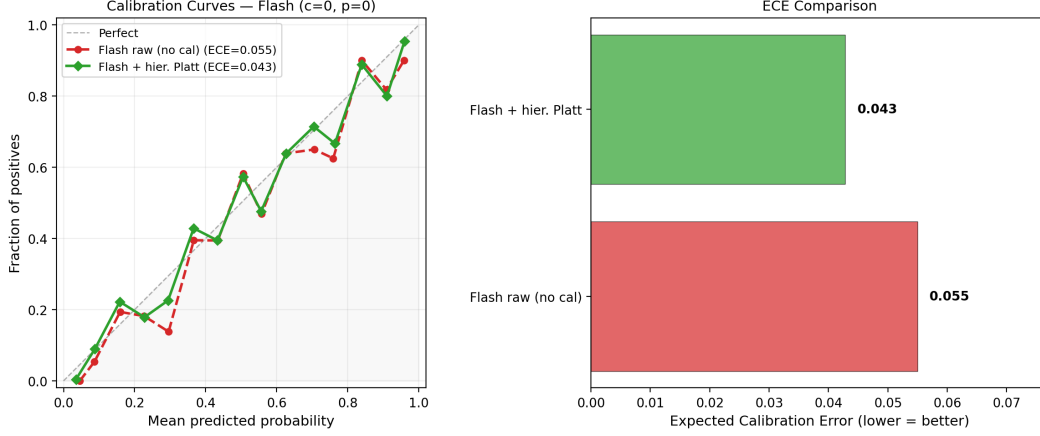


Figure 12: Reliability diagram for the Flash model on FB AUB ($n=791$ resolution dates). Hierarchical Platt scaling (green, $\text{ECE}=0.043$) improves on raw predictions (red, $\text{ECE}=0.055$). Right: ECE comparison.

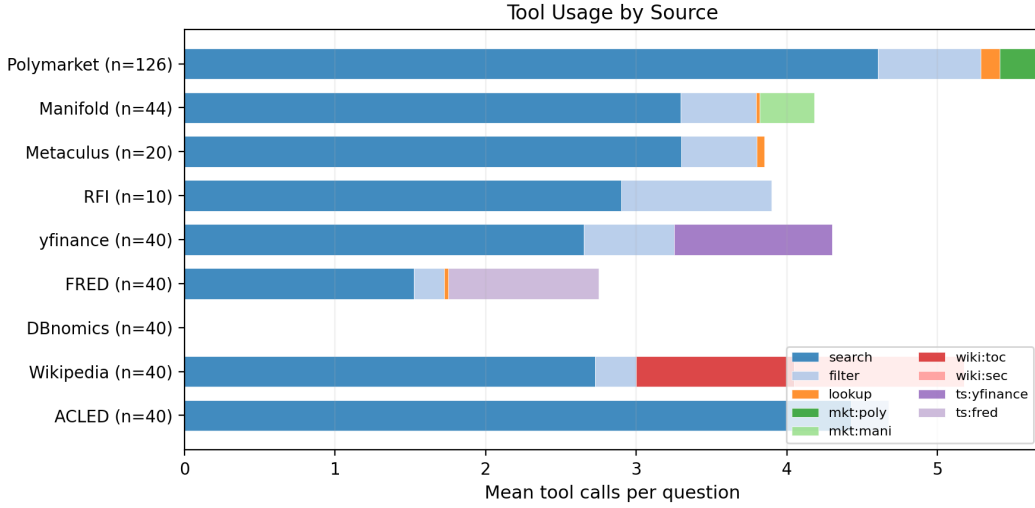


Figure 13: Mean tool calls per question by source for BLF (trial 1, AUB). Source-specific tools are used selectively. DBnomics bypasses the LLM (KNN model).

E.2 AIBQ2: Core Ablations

Table 16 shows the effects of ablating pieces of the core system. We use this to rank-order the importance of the system components, listed below. Note that, while the CIs are large, due to the small sample size, we replicate many of these findings in our much larger (and hence more reliable) FB experiments below.

- **Model capacity** is key: Flash scores only 32.3 MS (-7.2); Kimi scores 32.1 MS (-7.4).
- **Belief state** is the most important algorithmic component: *nobelief* drops to 35.1 MS (-4.4).
- **Search** is also critical: *zs* (no search) scores 36.7 MS (-2.8).
- **Thinking budget**: medium thinking level (instead of the default high) scores 37.5 MS (-2.0). However, this difference vanishes on FB (section G.2).

E.3 AIBQ2: Aggregation Ablations

In this section, we consider different ways to aggregate multiple trials. In particular we compare the simple mean in eq. (3) to the hierarchical shrinkage method using the standard-deviation estimator

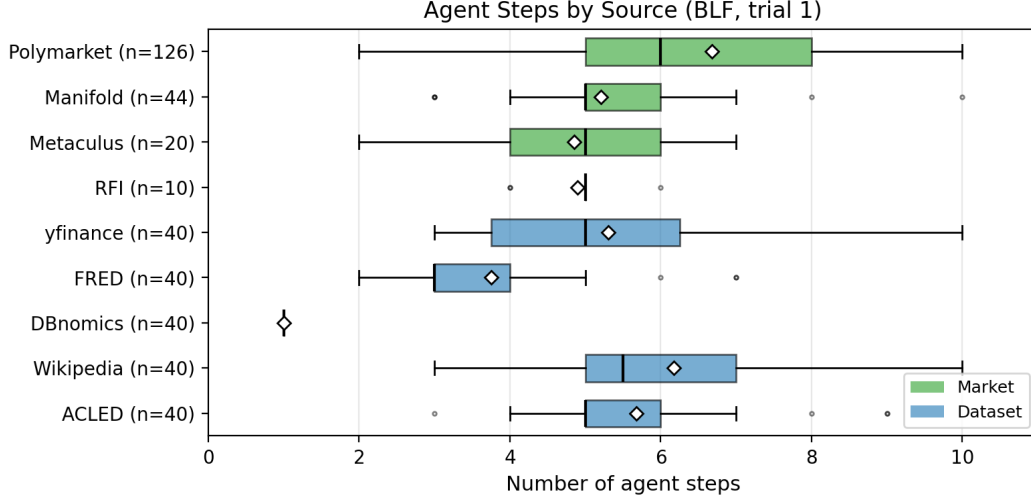


Figure 14: Agent steps per question by source (BLF, trial 1, AUB). Diamond = mean; box = IQR. Polymarket: mean 6.7 steps; DBnomics: 1 step (KNN bypass).

Table 16: AIBQ2 core ablations. BLF includes Platt calibration. Each row below the line changes one component (shown in bold). 95% bootstrap CIs in brackets.

Configuration	MBS $\uparrow$	BI $\uparrow$	BS $\downarrow$
BLF +cal (Pro/high/Brave)	39.5 [24,54]	63.1 [58,69]	.136 [.10,.18]
zs+cal (no search)	36.7 [21,51]	62.3 [57,68]	.142 [.10,.18]
medthink+cal (default think)	37.5 [22,52]	62.3 [57,68]	.142 [.11,.19]
kimi+cal (Kimi-K2)	32.1 [16,47]	60.8 [56,66]	.154 [.11,.20]
batch+cal (parallel)	36.1 [20,51]	62.0 [57,68]	.144 [.11,.19]
nobelief+cal (-belief)	35.1 [19,50]	61.9 [57,68]	.146 [.10,.19]
flash+cal (Flash)	32.3 [17,47]	60.9 [56,66]	.153 [.11,.19]

in eq. (8). The results are shown in fig. 15(a-c) and Table 17. We see that MBS and BS both improve significantly with more trials, and shrinkage aggregation slightly improves MBS over mean aggregation (though it hurts BI). For BI metric, things are more interesting. First we see that mean aggregation has no effect at all. Second we see that shrinkage aggregation hurts performance. See section C.8 for an explanation.

Table 17: AIBQ2 post-processing ablations. 95% bootstrap CIs in brackets.

Variant	MBS $\uparrow$	BI $\uparrow$	BS $\downarrow$
logit:5 (no cal)	32.7	61.3	.150
+cal (Platt, LOO-CV)	39.5 [24,54]	63.1 [58,69]	.136 [.10,.18]
median:5	33.5	62.0	.144
mean:3	33.2	61.5	.148
mean:1 (single trial)	26.4 [20,31]	60.0 [58,61]	.160 [.12,.21]

E.4 AIBQ2: Calibration Ablations

From Table 17, we see that calibration significantly improves MBS by 4.7, rising from 34.8 to 39.5, and improves BS by 0.008, dropping from 0.146 to 0.136. However, we see that calibration slightly hurts BI by 0.4, dropping it from 74.2 to 73.8. Note, however, that the CIs for all these estimates are wide, and on the much larger FB dataset in section D.5, we see that (hierarchical) calibration helps on all metrics.

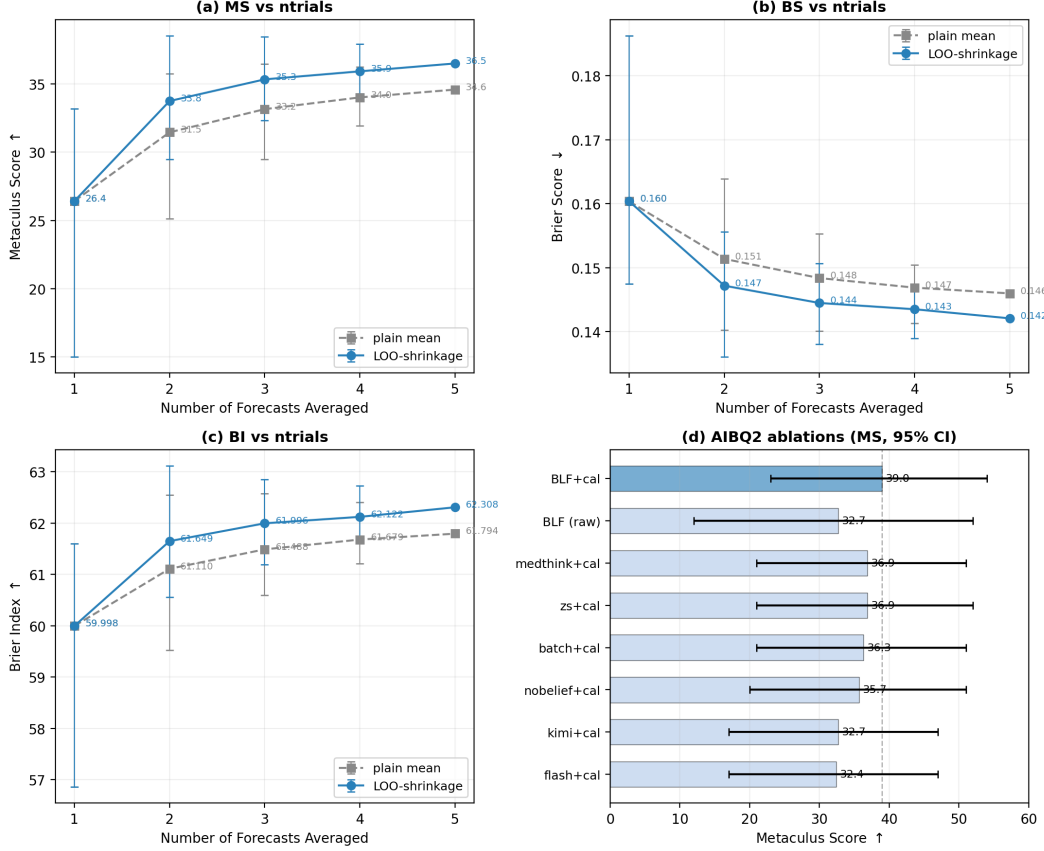


Figure 15: (a-c). Effect of increasing the number of trials on AIBQ2, as measured by MBS, BS and BI metrics. Error bars represent 95% CI over random trial subsets. We also compare plain mean (gray) vs std-shrinkage (blue). MBS and BS improve with more trials, but BI does not, for reasons explained in section C.8. (d) Effects of core ablations. Error bars represent the 95% bootstrap CIs shown in table 17.

F Ensemble Analysis

A common strategy for improving forecasts is to ensemble diverse models. We tested greedy ensembles of {Pro, Kimi-K2, Flash} (all without crowd, using mean aggregation of all trials from all members, uncalibrated).

Figure 16 shows MBS and BI as ensemble members are added. On ForecastBench, both metrics decline monotonically — adding weaker models hurts more than diversity helps. On AIBQ2, the picture is more nuanced: Pro+Kimi improves MS from 34.6 to 37.1 (Kimi brings genuinely different predictions, JSD=0.052), but BI declines in this case, suggesting the models are not sufficiently complementary. Adding Flash as a third member hurts MBS back down to 35.0.

The reason ensembling fails on FB is twofold: (1) the component models are highly correlated — they receive identical prompts, tools, and search results, differing only in the base LLM; (2) Kimi and Flash are individually weaker than Pro. Figure 17 shows the pairwise JSD: FB diversity is very low (0.006–0.014 bits) while AIBQ2 is moderately higher (0.028–0.052 bits), explaining the different ensemble behavior across datasets.

G ANOVA and Paired Analysis

G.1 Variance Decomposition

We perform a two-way ANOVA (method $\times$ question) on trial-level BI scores pooled across all 12 configurations in Table 2. Each observation is BI_{ijt} for method i , question j , trial t . The total sum of

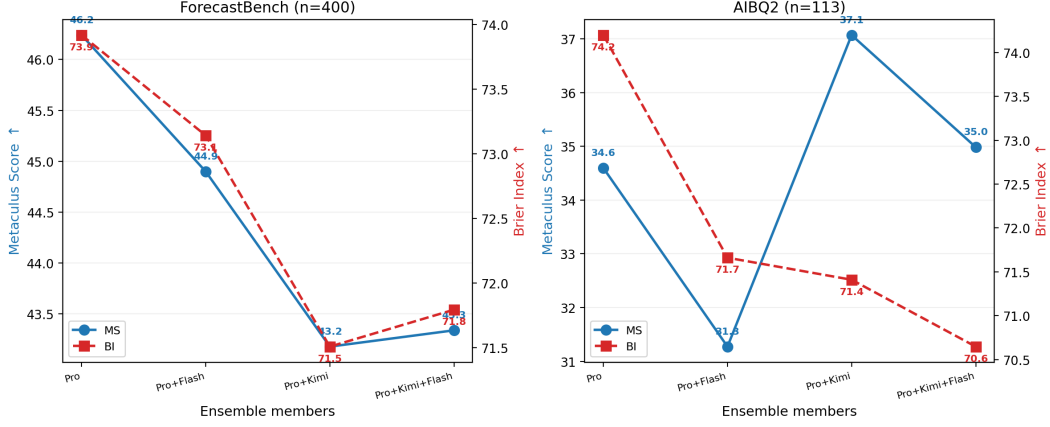


Figure 16: MBS and BI vs ensemble composition on FB (left) and AIBQ2 (right). On FB, both metrics decline with more members. On AIBQ2, MBS improves with Pro+Kimi (diversity helps) but BI declines (the models are not sufficiently complementary). All ensembles use uncalibrated forecasts.

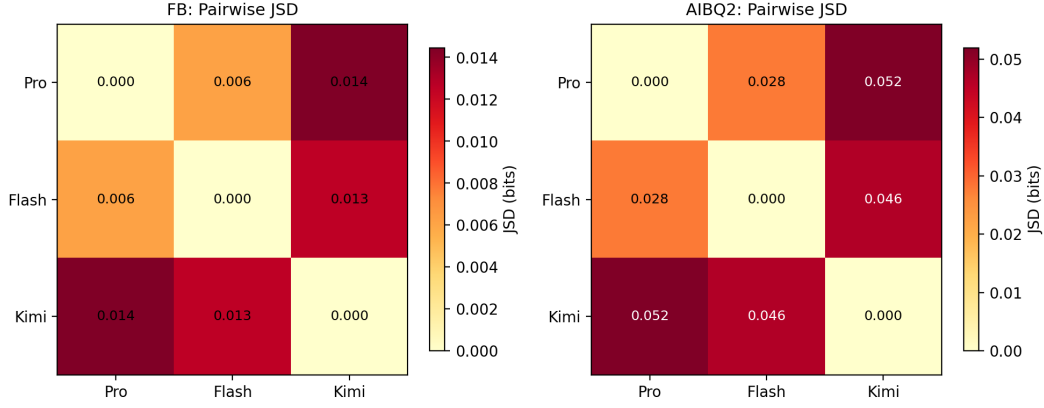


Figure 17: Pairwise Jensen-Shannon divergence between model forecasts on FB (left) and AIBQ2 (right). FB has very low diversity (JSD 0.006–0.014), explaining why ensembling does not help. AIBQ2 has higher diversity (0.028–0.052), especially Pro vs Kimi, enabling modest MBS improvement.

squares is decomposed as:

$$SS_{\text{total}} = SS_{\text{method}} + SS_{\text{question}} + SS_{\text{residual}}$$

where $SS_{\text{method}} = \sum_i n_i (\bar{y}_i - \bar{y})^2$ sums over the 12 methods, $SS_{\text{question}} = \sum_j n_j (\bar{y}_j - \bar{y})^2$ sums over the 400 questions, and the residual captures inter-trial variability.

The F -statistic tests whether the group means differ significantly: $F = MS_{\text{effect}}/MS_{\text{residual}}$, where $MS = SS/df$ is the mean square. A large F (relative to the F -distribution with appropriate degrees of freedom) indicates the effect is significant.

Table 18: Two-way ANOVA on trial-level BI scores (FB AUB, 12 methods $\times$ 400 questions $\times$ 5 trials).

Source	SS	% total	df	F
Method ($M=12$)	38.1	2.7%	11	164.5
Question ($Q=400$)	877.6	62.2%	399	104.3
Residual	494.3	35.1%	23,442	—
Total	1,410.1	100%	23,852	

From table 18, we see that question difficulty dominates (62.2% of total variance), consistent with the observation that some questions are inherently easy (e.g., “Will a vaccine for Sepsis exist by next

month?”) while others are genuinely hard (e.g., stock price movements), cf. [Frick et al., 2025]. See also fig. 18, which shows that on AIBQ2, most questions are very easy, so performance differences are dominated by a subset of hard questions.

Method effects account for only 2.7% of total variance, but are highly significant ($F = 164.5$, $p \ll 0.001$), confirming that the differences observed in Table 2 are real, not noise. The residual (35.1%) captures inter-trial variability within each (method, question) pair — i.e., the stochasticity of the LLM’s search and reasoning.

G.2 Paired Analysis

For the pairwise comparisons in Table 2, we use an additive two-way fixed-effects model:¹³

$$\text{BI}_{ij} = \mu + \alpha_i + \gamma_j + \epsilon_{ij} \quad (12)$$

where μ is the grand mean, α_i is the effect of method i (with $\sum_i \alpha_i = 0$), γ_j is the effect of question j (with $\sum_j \gamma_j = 0$), and ϵ_{ij} is the residual.

Estimation via alternating projections. We estimate $(\mu, \{\alpha_i\}, \{\gamma_j\})$ using an alternating least-squares (ALS) algorithm, which iterates:

$$\gamma_j \leftarrow \frac{1}{|\mathcal{I}_j|} \sum_{i \in \mathcal{I}_j} (\text{BI}_{ij} - \mu - \alpha_i) \quad (13)$$

$$\alpha_i \leftarrow \frac{1}{|\mathcal{J}_i|} \sum_{j \in \mathcal{J}_i} (\text{BI}_{ij} - \mu - \gamma_j) \quad (14)$$

where $\mathcal{I}_j$ is the set of methods that answered question j , and $\mathcal{J}_i$ is the set of questions answered by method i . This is equivalent to coordinate descent on the least-squares objective $\sum_{ij} (\text{BI}_{ij} - \mu - \alpha_i - \gamma_j)^2$, and converges to the unique global minimum since the objective is convex and separable in the two sets of parameters. In our setting, every method answers every question (balanced design), so ALS converges in a single iteration and produces the same estimates as direct computation: $\alpha_i = \bar{y}_{i\cdot} - \bar{y}$ and $\gamma_j = \bar{y}_{\cdot j} - \bar{y}$. However, the ALS formulation generalizes naturally to unbalanced designs (e.g., if some methods fail on some questions).

Pairwise comparisons. The pairwise Δ BI values in Table 2 are the estimated $\hat{\alpha}_i - \hat{\alpha}_{\text{ref}}$, which equal the paired mean differences $\frac{1}{n} \sum_j (\text{BI}_{ij} - \text{BI}_{\text{ref},j})$ since the question effects cancel. Bootstrap confidence intervals are computed by resampling questions (5,000 resamples). The p -value is computed as follows: for methods worse than the reference ($\Delta < 0$), p is the fraction of resamples where $\Delta \geq 0$ (i.e., the probability that the observed degradation is due to chance); for methods better than the reference ($\Delta > 0$), p is the fraction of resamples where $\Delta \leq 0$. We report significance using the convention $***p < 0.001$, $**p < 0.01$, $*p < 0.05$.

This approach is preferable to comparing raw means (Table 2) because it accounts for the fact that 62% of variance is due to question difficulty. Without pairing, a method that happens to be tested on easier questions might appear artificially good. By pairing on questions, the paired analysis isolates the true method effect and dramatically reduces the variance of the comparison.

H Analysis of the Datasets

H.1 AIBQ2

The AIBQ2 benchmark consists of 113 binary questions from the Metaculus AI Benchmark Tournament (Q2 2025). Questions were asked between 2025-04-22 and 2025-06-15, with resolution dates up to 2025-07-01. The base rate is 22% (25 True, 88 False). Figure 19 shows the forecast horizon distribution: most questions have short horizons (1–8 weeks), with many resolving on 2025-07-01

¹³In a true mixed-effects model, the question effects γ_j would be treated as random ($\gamma_j \sim \mathcal{N}(0, \sigma_q^2)$) and marginalized out when estimating the method effects α_i . This is statistically preferable (it accounts for uncertainty in γ_j and enables inference on new questions), but requires iterative algorithms such as REML. Since our design is balanced (every method answers every question), the fixed-effects estimates coincide with the mixed-effects estimates for α_i , so the simpler approach suffices.

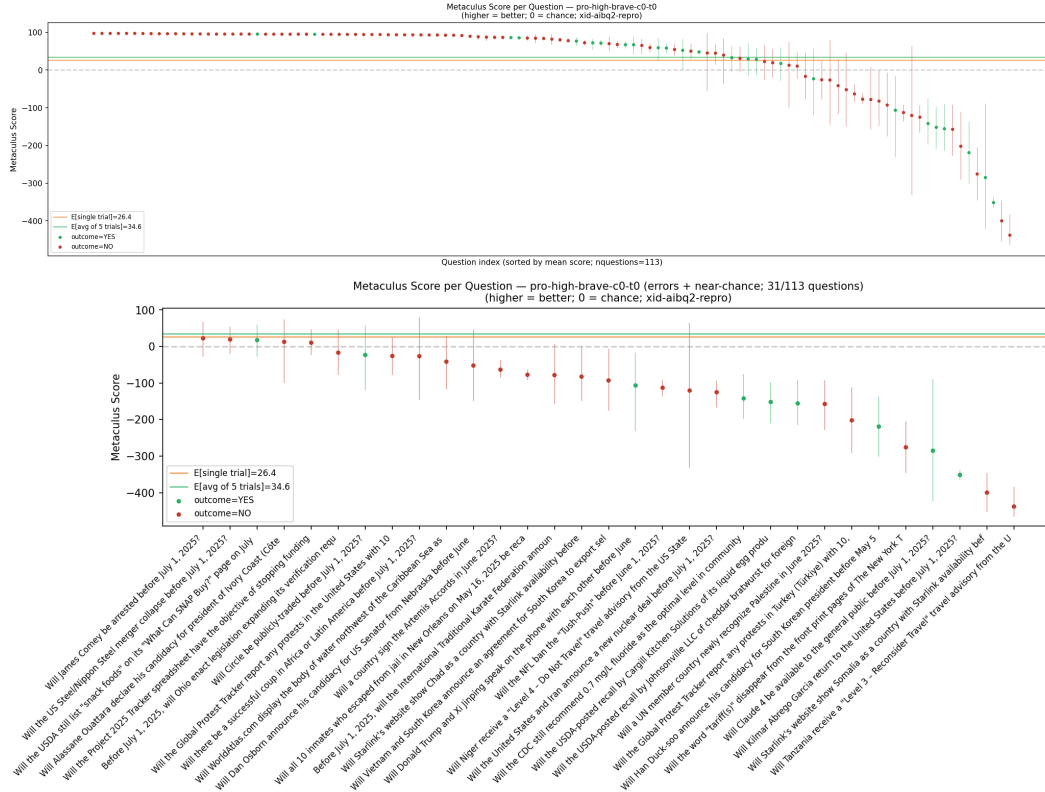


Figure 18: Top: MBS for all $n = 113$ AIBQ2 questions (most are easy). Bottom: Zoom into questions at or below chance. Error bars: bootstrap 95% CIs across 5 trials.

(the tournament deadline). See also fig. 20 for another visualization of this data. Table 19 shows some example questions, which we have clustered into 10 manually chosen topics. (The clustering is done by applying an LLM classifier to each question, and then grouping questions with the same class label or tag into the same cluster). We see that the most common topics are Domestic Politics (35/113) and Geopolitics & Conflict (29/113). By contrast, the FB topic distribution is quite different (see fig. 21).

Table 19: AIBQ2 topic distribution with example questions across 10 categories ($n = 113$). Domestic Politics dominates (35 questions).

Category	n	Example question
Domestic Politics	35	Will a state of emergency be in effect in Samoa on April 30, 2025?
Geopolitics & Conflict	29	Will there be a successful coup in Africa or Latin America before Jul...
Business & Industry	16	Will Uber be available in the Turks and Caicos Islands on July 1, 2025?
Society & Law	9	Will Harvard University lose its tax exempt status before July 1, 2025?
Macroeconomics	7	Will South Sudan ratify AfCFTA (the pan-African free trade agreement)...
Science & Technology	6	Will Starlink's website show Lesotho as a country with Starlink avail...
Health & Biology	5	Will the CDC raise the Travel Health Notice status of Colombia to abo...
Financial Markets	2	Will any of these companies get dropped from the Dow Jones Industrial...
Other	2	Will WorldAtlas.com display the body of water northwest of the Caribb...
Sports & Entertainment	2	Before July 1, 2025, will the International Traditional Karate Federa...
Total	113	

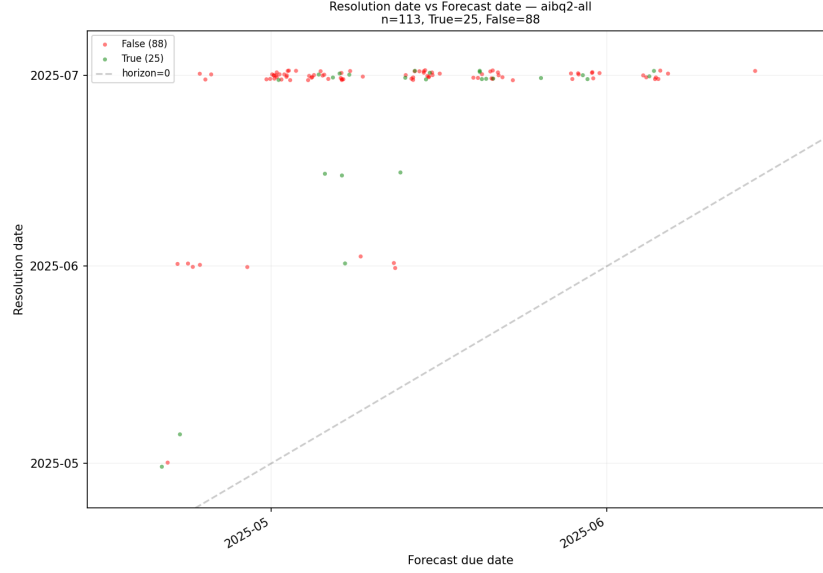


Figure 19: AIBQ2 date distribution of questions and answers. Green = resolved True (25), Red = resolved False (88).

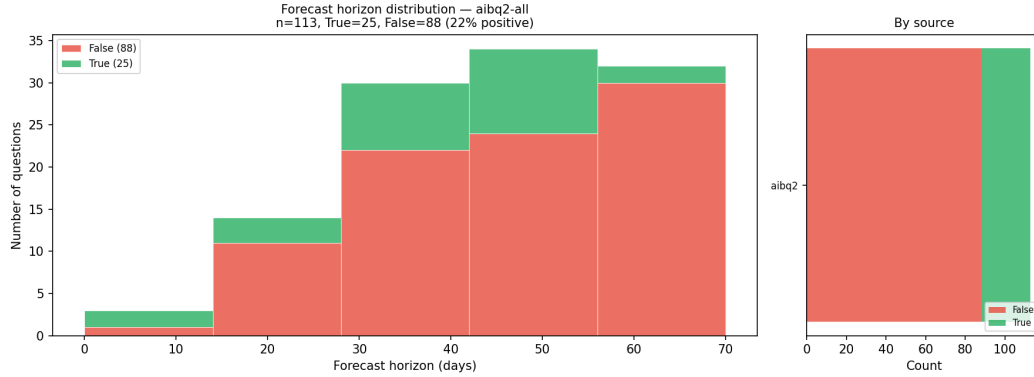


Figure 20: AIBQ2 forecast horizon distribution ($n = 113$). Right panel shows outcome balance.

H.2 ForecastBench (overall)

In this section, we discuss some properties of the FB dataset. Table 20 shows some key statistics. Figure 21 shows the topic distribution: Financial Markets (76) leads, followed by Sports & Entertainment (69) and Geopolitics & Conflict (66), with the remaining 7 categories more evenly distributed.

Figure 9 shows per-source BI for four methods. Performance of our best system (BLF+crowd+emp+cal) varies dramatically across sources: Wikipedia (83.0) and Manifold (90.7) are near-perfect, while yfinance (49.6) is essentially chance. FRED (53.2) is the second-weakest source.

We analyze each source in more detail below.

H.3 FB: Market sources

Market questions (Polymarket, Manifold, Metaculus, RFI) are judgemental forecasting questions similar in nature to AIBQ2 (see Table 19 for example topics), but spanning a wider range of sources and with a different topic distribution (Figure 21). Forecast horizons range from 1 week to 5 months (Figure 22).

Table 20: Question composition by source for FB Tranches A and B. “BR” = base rate (fraction resolving True). Market sources provide judgemental forecasting questions; dataset sources provide univariate time-series questions with multiple resolution dates per question. ACLED = Armed Conflict Location and Event Data. FRED = Federal Reserve Economic Data. DBnomics = database of international economic statistics.

Source	URL	n_A	n_B	n	BR(A)	BR(B)
RFI	randforecastinginitiative.org	5	5	10	0.00	0.00
Manifold	manifold.markets	18	26	44	0.22	0.15
Metaculus	metaculus.com	10	10	20	0.10	0.20
Polymarket	polymarket.com	67	59	126	0.13	0.08
<i>Market total</i>		100	100	200	0.14	0.11
ACLED	acleddata.com	20	20	40	0.25	0.00
DBnomics	db.nomics.world	20	20	40	0.95	0.80
FRED	fred.stlouisfed.org	20	20	40	0.60	0.55
Wikipedia	wikipedia.org	20	20	40	0.20	0.25
Yahoo Finance	finance.yahoo.com	20	20	40	0.20	0.45
<i>Dataset total</i>		100	100	200	0.44	0.41
Total		200	200	400	0.29	0.26

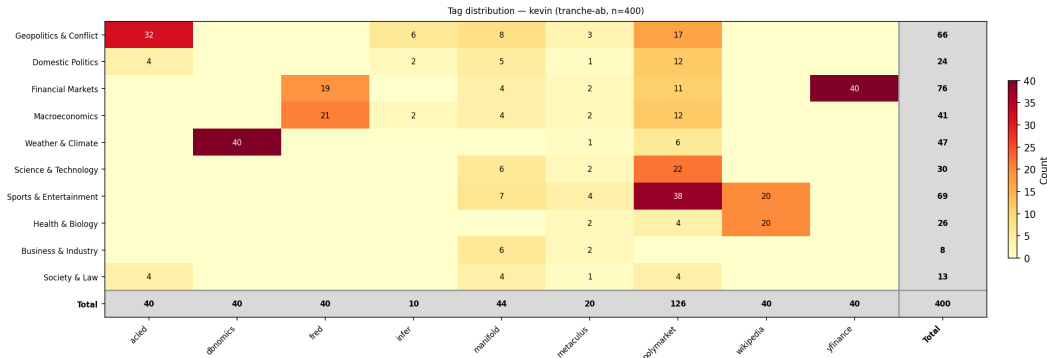


Figure 21: FB AUB topic distribution across 9 sources.

The base rates are very low (Table 20): Polymarket 11%, Manifold 18%, Metaculus 15%, RFI 0%. This means most questions resolve “No.”. Furthermore, these questions come with a market estimate; using this as the predicted probability achieves high market BI — a strong baseline that is very hard to beat. Our system’s main advantage on market questions comes from calibration, which sharpens the already-good raw forecasts (Table 2).

H.4 FB: Dataset sources

Dataset questions are univariate time series forecasting problems, which are converted to binary form by asking whether (some function of) the value of a quantity of interest at a future resolution date will be higher or lower than the value at the forecast date. Topics are very narrow, and depend on the source, as shown in Figure 21. Forecast horizons are fixed at 7, 30, and 90 days, as shown in Figure 23. We give more details on individual dataset sources below.

H.5 FB: Yahoo Finance (stock prices)

Yahoo Finance questions ask whether a stock’s closing price on a future resolution date will be higher than its closing price on the forecast due date. Each question has 8 resolution dates ranging from 1 week to 10 years out. Figure 24 shows an example.

The question text uses placeholders {resolution_date} and {forecast_due_date} that are substituted at preprocessing time. Crucially, the question does *not* specify the reference price on the forecast due date — it only provides a freeze_datetime_value from an earlier date (when the

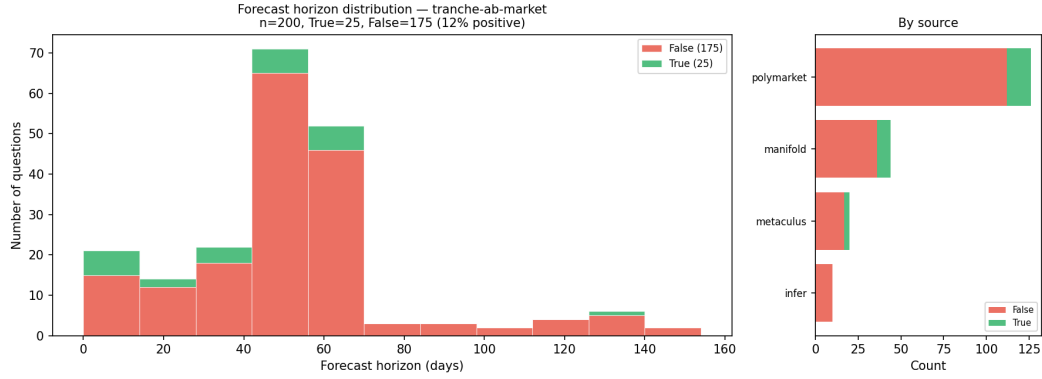


Figure 22: FB market questions: forecast horizon distribution ($n = 200$). Most horizons are 1–5 months. Base rate is 12% (heavily skewed toward False).

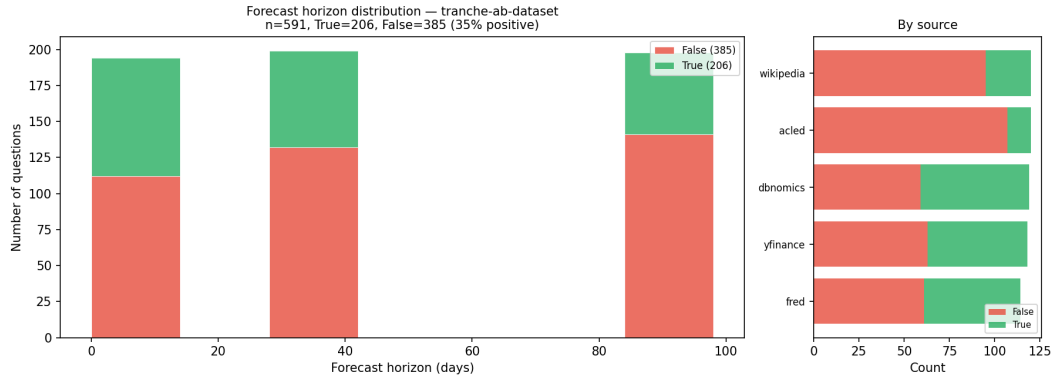


Figure 23: FB dataset questions: forecast horizon distribution ($n = 200$ questions, 591 resolution dates). Horizons are fixed at 7, 30, and 90 days (only the first 3 of 8 standard resolution dates have resolved by our evaluation cutoff). Base rate is 35%.

question was created, typically 10 days before the forecast due date). For example, the CPB question below has `freeze_datetime_value = 30.15` from 2025-10-16, but the actual closing price on the forecast due date (2025-10-24, since 10-26 is a Sunday) was 30.79. Our system retrieves the correct reference value via the `fetch_ts_yfinance` tool, which downloads the price history and returns the closing price on or before the forecast due date.

Figure 24 illustrates why yfinance questions are inherently difficult: stock prices follow an approximate random walk, making it nearly impossible to predict whether the price will be higher or lower than the reference value on any given future date. The optimal strategy is to predict $p \approx 0.5$, which our system correctly learns (yfinance BI ≈ 0.5 across both tranches; see Table 20).

Example question (verbatim, abbreviated).

```
{
  "forecast_due_date": "2025-10-26", // global
  "id": "CPB",
  "source": "yfinance",
  "question": "Will CPB's market close price on
    {resolution_date} be higher than its market
    close price on {forecast_due_date}?",
  "resolution_criteria": "Resolves to the market
    close price at https://finance.yahoo.com/quote/CPB.
    If the resolution date coincides with a day the
    market is closed, the previous close is used.",
```

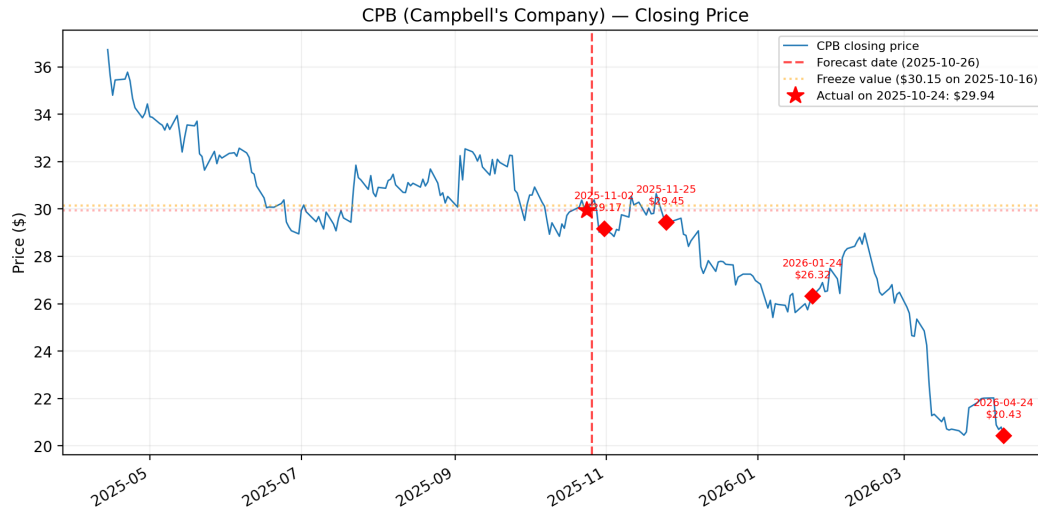


Figure 24: CPB (Campbell’s Company) closing price. Red dashed line: forecast date (2025-10-26). Star: actual price on 2025-10-24 (\$29.94, since 10-26 was Sunday). Orange: freeze value (\$30.15 from 10-16). Red diamonds: resolution date prices (all below reference, all resolved False). The decline to ~\$20 was unpredictable at forecast time.

```

"freeze_datetime": "2025-10-16T00:00:00+00:00",
"freeze_datetime_value": "30.15",
"resolution_dates": ["2025-11-02", "2025-11-25",
    "2026-01-24", ...],
"resolved_to": [0, 0, 0, ...] // all False
}

```

H.6 FB: FRED (economic indicators)

FRED (Federal Reserve Economic Data) questions ask whether an economic indicator will have increased by each resolution date compared to the forecast due date. An example question is shown below:

Question (DTB6): “Will the Federal Reserve’s 6-month secondary market treasury bill rate have increased by {resolution_date} compared to its value on {forecast_due_date}?”

Freeze value: 3.69% (as of 2025-10-16)

Resolution dates: 2025-11-02, 2025-11-25, 2026-01-24, 2026-04-24, 2026-10-26, 2028-10-25, 2030-10-25, 2035-10-24

The 36 series in our tranches span interest rates (DTB6, DGS30), corporate bond spreads (BAMLC0A0CM), exchange rates (DEXUSUK), stock indices (DJIA), and banking data (DP-SACBW027SBOG).

Unlike yfinance (approximate random walk), many FRED series exhibit persistent trends. For example, Figure 25 shows the 6-month T-bill rate declining steadily from ~5.25% to ~3.5% over 2024–2026, reflecting Fed rate cuts. The first resolution date (2025-11-02) caught a brief uptick (resolved True), but subsequent dates resolved False as the rate continued falling. This trend is partially predictable from recent history, giving our `fetch_ts_fred` tool useful context.

Figure 26 shows the USD/GBP exchange rate, which is harder to predict — more similar to a random walk, with a mix of True and False resolutions.

Overall, FRED has mixed predictability: of our 36 series, roughly 8 exhibit strong autocorrelation (trending), 18 are effectively random walks, and the rest are intermediate. Trend-following series (interest rates, banking aggregates) reward tools that detect recent direction, while noisy series

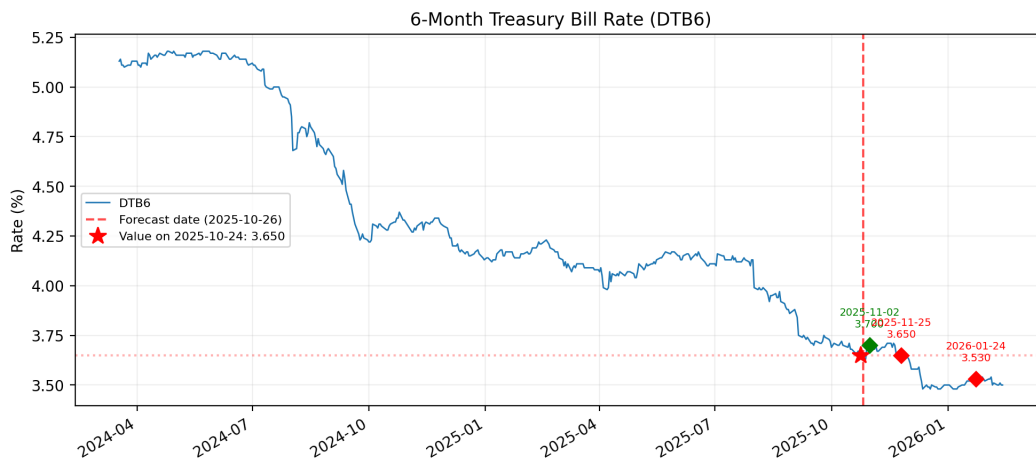


Figure 25: 6-Month Treasury Bill Rate (DTB6). Forecast date is 2025-10-26 (Sunday), so the reference value is from 2025-10-24 (red star). Clear downward trend from Fed rate cuts. Green diamond = resolved True (rate increased); red = resolved False.

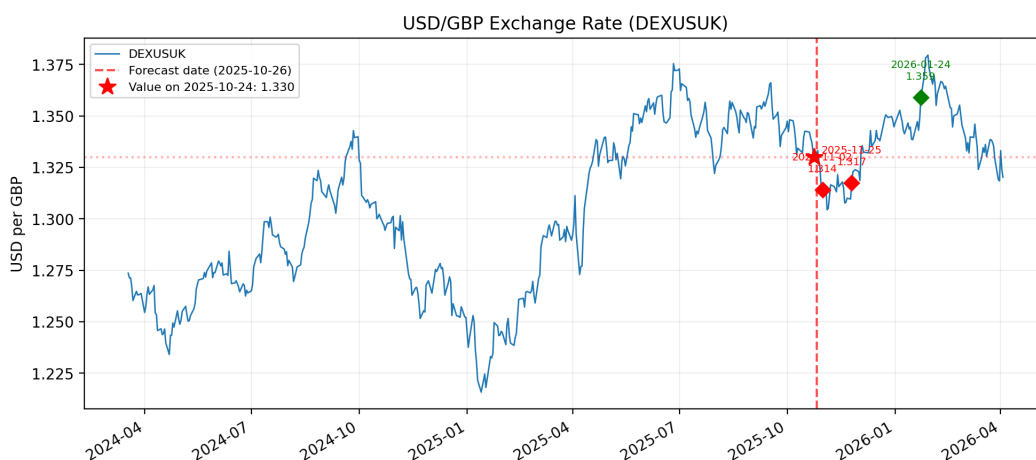


Figure 26: USD/GBP exchange rate (DEXUSUK). Reference value from 2025-10-24 (red star, since forecast date 2025-10-26 is a Sunday). More erratic than interest rates — harder to predict.

(exchange rates, bond yields) are closer to chance. The combined base rate for FRED is 57% (Table 20), between DBnomics (88%) and yfinance (33%).

A natural improvement would be to classify each FRED series into a small number of patterns (random walk, trending, mean-reverting) and apply the appropriate statistical model — e.g., $p = 0.5$ for random walks, a trend-extrapolation model for persistent series. ForecastBench’s full question bank contains 165 unique FRED series, so this per-series approach would require a lightweight classifier, which we leave to future work.

H.7 FB: DBnomics (temperature)

All DBnomics questions in ForecastBench ask whether the daily average temperature at a French weather station will be higher on a future resolution date than on the forecast due date. Our tranches contain 29 stations (24 metropolitan France, 5 overseas), as shown in Figure 27; each generates one question per forecast date with 8 resolution dates. Figure 28 shows ~14 years of daily temperature data for one station (Mont-de-Marsan), exhibiting strong annual seasonality.

Although the stations are spatially distributed, we found that pooling bias-corrected observations from nearby stations does not improve forecasts (negligible BI gain). This is because each station

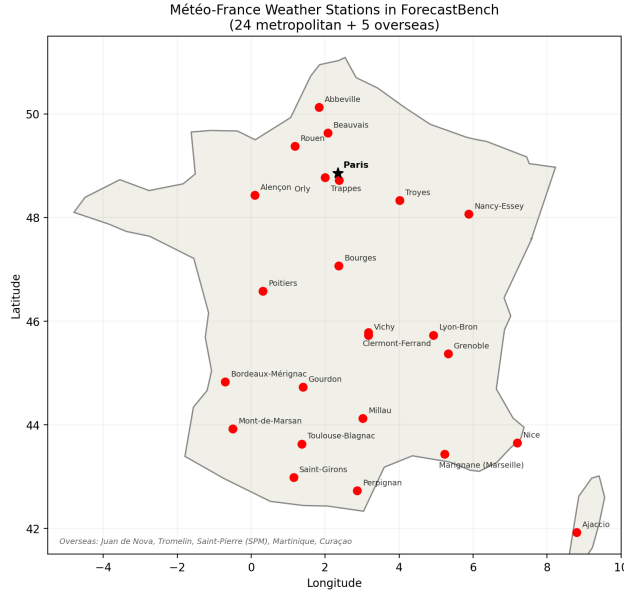


Figure 27: Météo-France weather stations used in ForecastBench. Stations are highly correlated (mean pairwise $r = 0.93$).

already has ~ 200 historical observations per resolution date, so the per-station empirical exceedance estimate is already well-determined. The main bottleneck for dbnomics is near-horizon weather forecast data (e.g., 7-day forecasts), not statistical estimation from historical data.

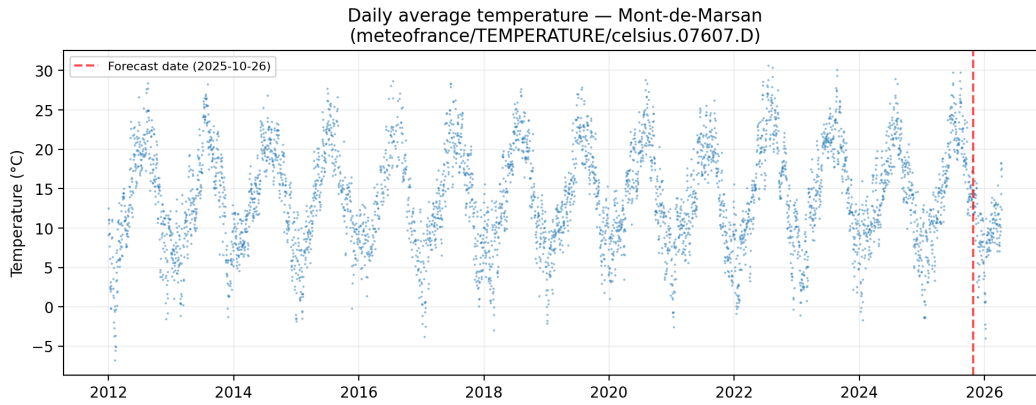


Figure 28: Daily average temperature at Mont-de-Marsan, 2012–2025. Strong annual seasonality makes statistical forecasting effective.

Figure 29 zooms into the period around the forecast date. The key challenge is the same as yfinance: the question provides a `freeze_datetime_value` of 14.4°C from 2025-10-16, but the actual value on the forecast date (2025-10-26) was 11.7°C — the threshold the agent must predict against. Our `fetch_ts_dbnomics` tool retrieves the correct value and computes an empirical exceedance probability (see Appendix I).

Unlike yfinance (random walk), temperature has strong seasonal structure: the October-to-January decline is predictable from ~ 14 years of historical data. Our empirical exceedance model achieves good BI by computing $P(\text{temp} > \text{threshold})$ from same-calendar-date observations in prior years (see section I for model comparison). We bypass the LLM entirely for DBnomics, since LLM reasoning does not improve on the statistical model. Our system outperforms Cassi on this source (BI 60.8 vs Cassi 51.0). Further improvement likely requires short-term weather forecast data for near-horizon resolution dates.

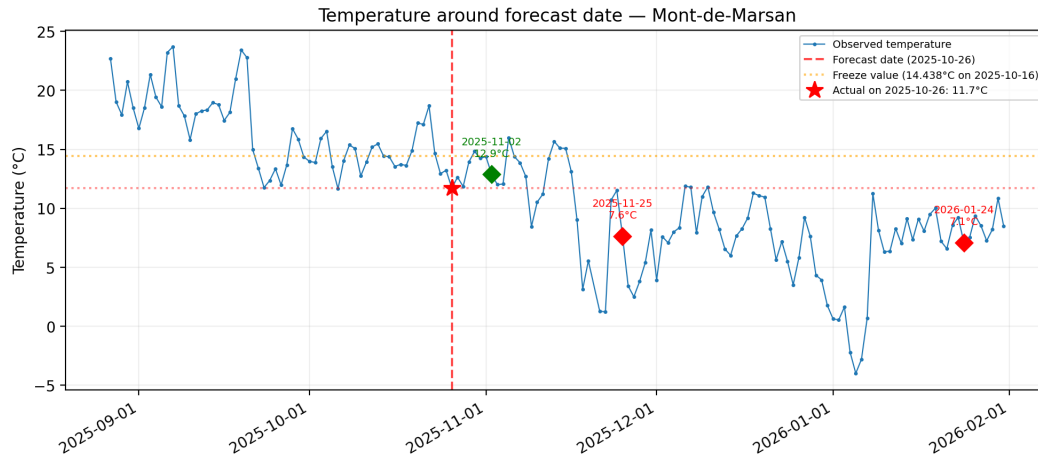


Figure 29: Temperature around the forecast date for Mont-de-Marsan. Star: actual value on forecast date (11.7°C). Orange: stale freeze value (14.4°C). Green diamond: resolution date warmer than reference; red: colder. The seasonal decline from October to January is predictable from historical data.

Example question (verbatim, abbreviated).

```
{
  "forecast_due_date": "2025-10-26", // global
  "id": "meteofrance_TEMPERATURE_celsius.07607.D",
  "source": "dbnomics",
  "question": "What is the probability that the
    daily average temperature at Mont-de-Marsan
    will be higher on {resolution_date} than on
    {forecast_due_date}?",
  "url": "https://db.nomics.world/meteofrance/
    TEMPERATURE/celsius.07607.D",
  "freeze_datetime": "2025-10-16T00:00:00+00:00",
  "freeze_datetime_value": "14.438",
  "resolution_dates": ["2025-11-02", "2025-11-25",
    "2026-01-24", ...],
  "resolved_to": [1, 0, 0, ...] // True, False, False
}
```

H.8 FB: ACLED (armed conflict)

ACLED (Armed Conflict Location & Event Data) questions ask whether the number of conflict events of a given type in a given country will exceed a threshold over a future 30-day window. There are two question templates (Figure 30):

- **Any increase** (20 questions): will events exceed the baseline level? These have a 22% base rate. Example:
Any increase: “Will there be more ‘Protests’ in Sri Lanka for the 30 days before {resolution_date} compared to the 30-day average of ‘Protests’ over the 360 days preceding 2025-11-09?”
 Resolution dates: 2025-11-16, 2025-12-09, 2026-02-07.
 Resolved: False, False, True (protest activity surged later).
- **10x spike** (20 questions): will events in the next 30 days exceed *ten times* the 360-day rolling average? These have a 0% base rate — no country in our tranches experienced such an extreme spike. Example:
10x spike: “Will there be more than ten times as many fatalities in Finland for the 30 days before {resolution_date} compared to *one plus* the 30-day average of fatalities over the 360 days preceding 2025-11-09?”

The “one plus” ensures the threshold is non-zero even when the baseline average is 0.
Resolved False on all 3 dates.

Questions span 30 countries (from Antarctica to Venezuela) and 6 event types (Battles, Strategic developments, Riots, Violence against civilians, Explosions/Remote violence, Protests). No source-specific data tool is available for ACLED (the API requires special access), so the agent relies entirely on web search.

The very low overall base rate (11%) means that predicting “No” for all questions is a strong baseline. The 10x-spike questions are particularly easy: since none resolved True, any well-calibrated system should predict near 0 for these. The agent’s main challenge is the “any increase” questions, which require understanding recent conflict dynamics via web search.

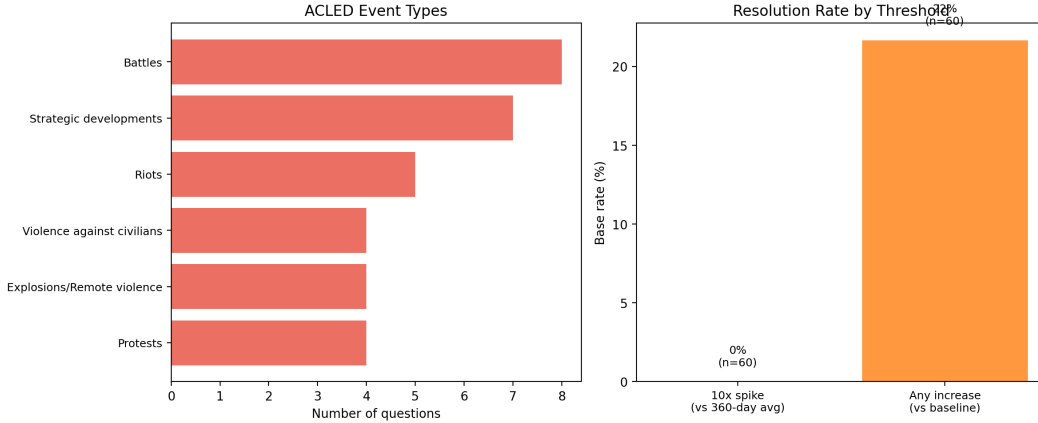


Figure 30: ACLED questions in AUB. Left: event type distribution. Right: base rate by threshold type. The “10x spike” questions never resolved True.

H.9 FB: Wikipedia

Wikipedia questions ask whether specific facts on a Wikipedia page will change by the resolution date. Our 40 questions come from three articles:

- **List of infectious diseases** (20 questions, BR=0%): “Will a vaccine have been developed for [disease] by [resolution date]?” Since vaccine development takes years, none of these resolved True over our 1–3 month horizons.
- **FIDE rankings** (15 questions, BR=22%): either “Will [player]’s FIDE ranking be as high or higher?” (5 rank questions) or “Will [player]’s Elo rating be at least 1% higher?” (10 Elo questions). The rank questions are easier (3/5 always True), while gaining 1% Elo in 1–3 months is very rare (0/10 True). See Figure 31.
- **List of world records in swimming** (5 questions, BR=100%): “Will [swimmer] still hold the world record for [event]?” All resolved True (world records are rarely broken in a few months).

The overall base rate is 21%, but it varies dramatically by article (0%, 22%, 100%). Our system achieves very high BI on Wikipedia (83.0; see Figure 9) because the questions are often answerable from the structure of the question itself: vaccine questions are almost certainly No, world record questions are almost certainly Yes. The `fetch_wikipedia_toc/section` tools allow the agent to verify these priors by checking the actual Wikipedia page content.

I Time-Series Forecasting Models

In this section we discuss specialized statistical modeling tools which we optionally make available to the agent for certain data sources. These all return an estimate of $\hat{p}_q^r$ defined in eq. (1). These are

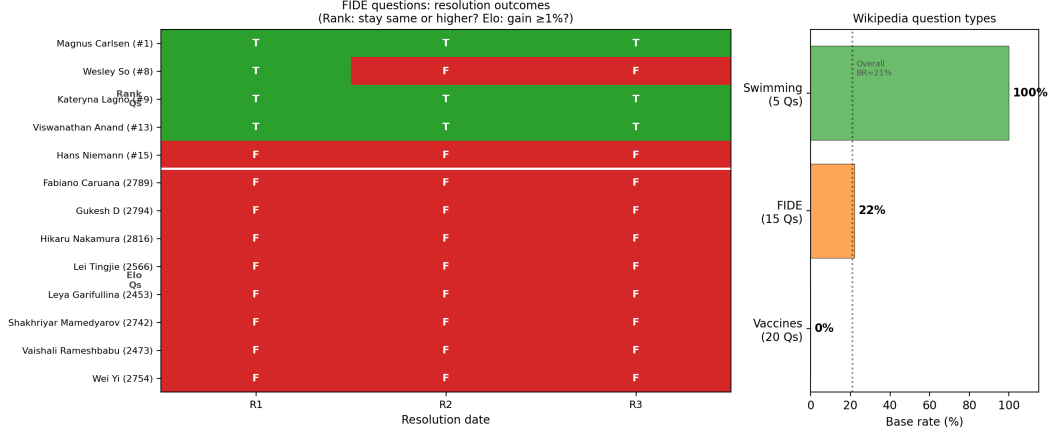


Figure 31: Left: FIDE question outcomes. Rank questions (top 5) mostly resolve True; Elo $\geq 1\%$ questions (bottom 10) all resolve False. Wesley So (#8) is the only mixed case (True, False, False — his ranking briefly improved then dropped). Right: base rates for the three Wikipedia question types.

all simple heuristic estimators that we developed to see if they helped the LLM with its task. Clearly many more sophisticated methods could be used, but that is not the focus of this paper.

I.1 KNN model for DBnomics

Table 21: Time-series model comparison on DBnomics questions (AUB, $n=40$ questions, 119 resolution dates). All models above the line are evaluated standalone (no LLM). KNN is our default for DBnomics and outperforms Cassi.

Model	BI
KNN (± 10 days, 10+ years)	59.1
Always 0.5 (baseline)	50.0
Cassi (reference)	51.0

We tried several simple models for estimating $\hat{p}_q^r$ for DBnomics data. The best performing method, which we call KNN, is a non-parametric estimator which is analogous to a binary k -nearest-neighbor classifier, where “neighbors” are selected by day-of-year proximity. More precisely, define the neighbor set as follows:

$$\mathcal{N}_{a,w}(q, r) = \{y_q(t_i) : |\text{doy}(t_i) - \text{doy}(r)| \leq w, t_i \geq f_q - a\}, \quad (15)$$

where $\text{doy}(\cdot)$ extracts the day-of-year, a is the maximum age (to encourage focusing on data close to q ’s forecast date), w is the window size around $\text{doy}(r)$, and i indexes across all available data. The forecast is the Laplace-smoothed empirical exceedance frequency:

$$\hat{p}_q^{\text{knn}}(a, w) = \frac{|\{y_q(t_i) \in \mathcal{N}_{a,w}(q, r) : y_q(t_i) > v_q\}| + 1}{|\mathcal{N}_{a,w}(q, r)| + 2}. \quad (16)$$

This directly estimates eq. (1) without distributional assumptions. For DBnomics, we use a window of $w = 10$ days, and set $a = \infty$ to remove age restrictions, reflecting an assumption the periodic distribution is stationary. With ~ 14 years of daily data, each resolution date typically has $|\mathcal{N}_{a,w}(q, r)| \approx 200$ neighbors.

Table 21 shows results on AUB. Our KNN method achieves BI 59.1, outperforming Cassi (51.0) on this source. We also tried feeding the time series directly to the LLM, as well as linear and harmonic models, but KNN was the best standalone approach. Further improvement might come from short-term weather forecast data for near-horizon resolution dates.

I.2 Linear trend models for yfinance and FRED

For non-seasonal time series, such as yfinance and FRED, we use a parametric approach, which combines a local linear model with the baseline estimate of 0.5:

$$p_q^{\text{safe-linear}}(\alpha) = \alpha \cdot p_q^{\text{linear}} + (1 - \alpha) \cdot 0.5 \quad (17)$$

where α is a source-specific shrinkage factor controlling how much to trust the linear extrapolation. We set $\alpha = 0.1$ for yfinance, which reflects that stock prices follow an approximate random walk, so the linear extrapolation is unreliable. We set $\alpha = 0.5$ for FRED. (All α values were hand-tuned based on informal experimentation; systematic cross-validation is left to future work.)

The linear probability estimate is given by

$$p_q^{\text{linear}} = \Phi^c\left(\frac{v_q - \hat{y}(r)}{\hat{\sigma}}\right) \quad (18)$$

where Φ^c is the standard normal survival function, and $\hat{y}(t) = at + b$ is the linear model fit by OLS on the most recent W observations, $\mathcal{H}(s_q, f_q; W)$. (We use $W = 30$ for FRED and $W = 60$ for yfinance.) $\hat{\sigma}$ is the standard deviation of the residuals for this data source s_q .

I.3 Hybrid KNN and linear trend model for yfinance and FRED

For FRED and yfinance, we also consider a hybrid parametric and non-parametric estimate of the form

$$\hat{p}_q^{\text{hybrid}}(\alpha, \beta) = \beta \cdot p_q^{\text{safe-linear}}(\alpha) + (1 - \beta) \cdot p_q^{\text{prior-year}} \quad (19)$$

where $p_q^{\text{prior-year}} = p_q^{\text{knn}}(a, w)$ uses the same nonparametric exceedance estimate as eq. (16), but with a narrower window ($w = 7$ days) and limited to the 5 most recent years, by setting $a = 5 \times 365$ days. This captures annual periodicity: for a given resolution date, we check whether the value exceeded the threshold at the same time of year in prior years. Expanding out, this estimate combines 3 terms:

$$\hat{p}_q^{\text{hybrid}}(\alpha, \beta) = \alpha\beta p_q^{\text{linear}} + (1 - \beta) p_q^{\text{prior-year}} + (1 - \alpha)\beta \cdot 0.5 \quad (20)$$

We set $\beta = 0.5$ for both FRED and yfinance when prior-year data is available ($\beta = 1$ otherwise, falling back to $p^{\text{safe-linear}}$ alone). For DBnomics, the hybrid model is not used — we apply the KNN model (eq. (16)) directly.

J Prompts

This appendix lists the prompts used in our system and the zero-shot baseline.

J.1 System Prompt

The following is the system prompt used for all BLF configurations (abbreviated for space; source-specific tool descriptions omitted).

```
You are an expert superforecaster. Your task is to predict the
probability that a binary question will resolve to YES given
information up to a certain date.

You work in a tool-use loop:
1. Read the question, its resolution criteria, and background.
2. Form a base rate estimate (outside view / reference class
reasoning).
3. Perform a loop:
3a. Choose ONE tool to call.
3b. After each tool call, your belief state is updated.
4. When you have gathered enough evidence, call submit(probability,
reasoning).

Belief state rules:
- Evidence lists should ACCUMULATE across steps.
```

- Each evidence item MUST cite its source (search_X_result_Y).
 - Include update_reasoning explaining WHY evidence changed your probability.
 - Consider RECENCY and AUTHORITATIVENESS of sources.
- Rules:
- You MUST call submit before step {max_steps}.
 - Call submit once probability has stabilized.
 - Probabilities must be between 0.05 and 0.95.

J.2 Question Prompt

Each question is formatted as follows (with crowd/prior sections included only when crowd=1 or emp=1):

```
# Question
{question text}

## Background and resolution criteria
{background}
{resolution_criteria}

## Resolution dates
{dates}
You must submit {n} probabilities (one per resolution date). Your
uncertainty should INCREASE with forecast horizon.

## Knowledge cutoff
{cutoff_date}
You must not use any information from after this date.

## Market estimate (if crowd=1, market questions only)
The market estimate on {date} was {value}.

## Prior estimate (if emp=1, dataset questions only)
{explanation}: {prior}.
Use this as your starting point, but adjust based on
question-specific evidence from search and tools.
```

J.3 Zero-Shot Prompt

When search=none and tools=0 (zero-shot setting), we use a simplified prompt based on Halawi et al. [2024], which is also used by ForecastBench for its zero-shot leaderboard entries:

```
You are an expert superforecaster, familiar with the work of Tetlock
and others. Make a prediction of the probability that the question
will be resolved as true. You MUST give a probability estimate
between 0 and 1 UNDER ALL CIRCUMSTANCES. If for some reason you
can't answer, pick the base rate, but return a number between 0 and
1.

Question: {question}
Question Background: {background}
Resolution Criteria: {resolution_criteria}
Question close date: {close_date}
[If freeze value available:] The freeze value is {value}. {explanation}

Output your answer (a number between 0 and 1) with an asterisk at
the beginning and end of the decimal. Do not output anything else.
Answer:
{{ Insert answer here }}
```

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